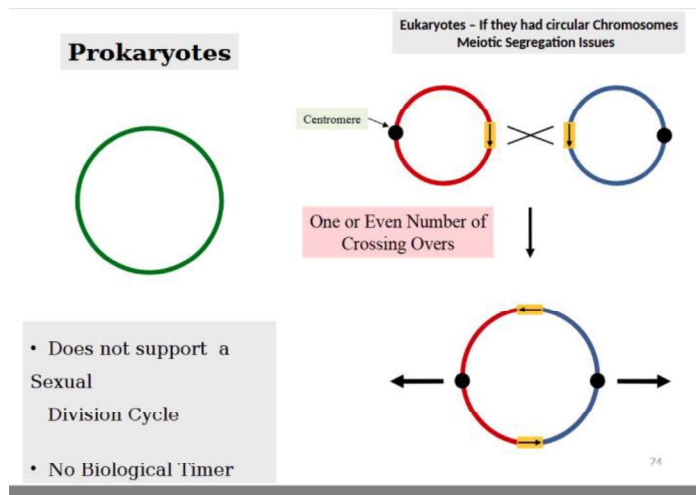


“The price of having recombinant cells means we do not live forever, but this is good as this stops the high risk of cancer mutations and alike”

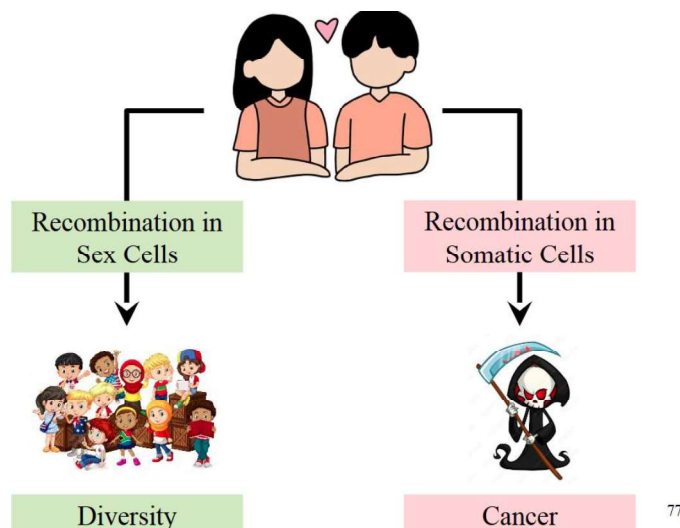
Organization of a Genome – Why Eukaryotes have Linear Chromosome:



It is very difficult to come up with methods for crossing over with circular chromosomes than it is with linear chromosomes which is why nature/evolution has favored this.

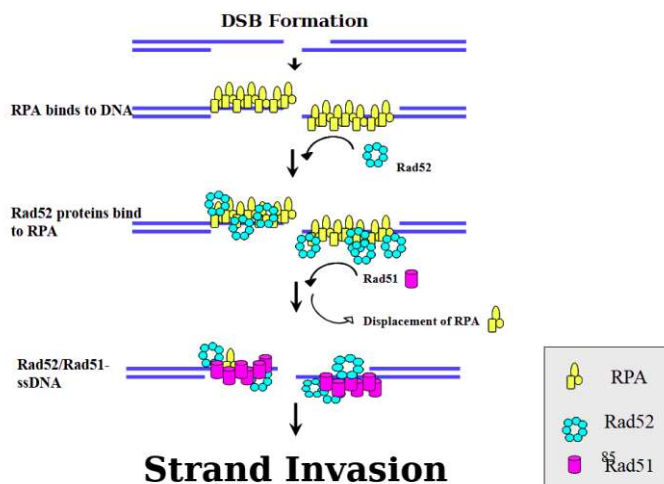
• Lecture 4: Mitotic DNA DSB Repair

Recombination is not always desirable:



Instead of the crossover happening in the repairing model like in meiosis it will for mitosis have helicase activity fix it, no unnecessary experiments in our cells.

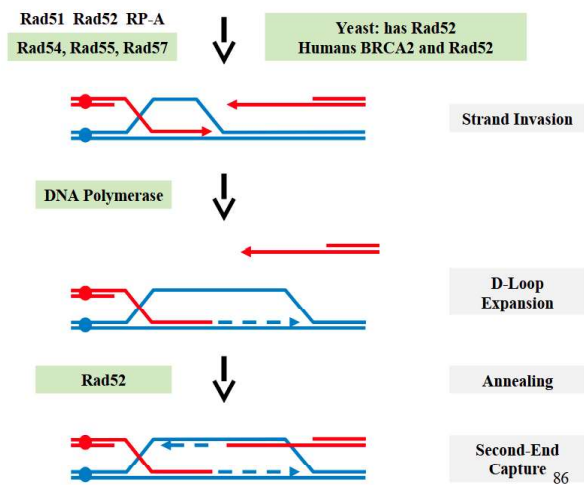
Eukaryotic Core Proteins involved in Homologous Recombination:



The cell does not know that this is a recombination fork so RPA: Is it a recombination fork or a lesion. (as a kind of inhibitor)

The RPA acts as a recruiter of rad52 which is a handing over enzymes that strips off RPA and load Rad51 as a safety measure.

Eukaryotic Core Proteins involved in Homologous Recombination: mitosis:



Annealing is promoted by the yeast Rad52 protein.

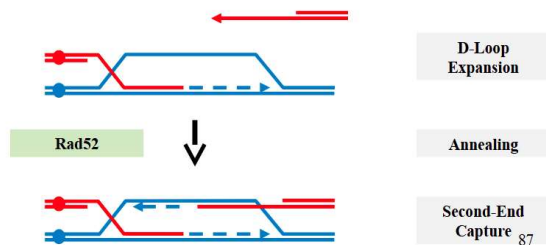
DNA strand annealing is promoted by the yeast Rad52 protein

(genetic recombination/repair/*Saccharomyces cerevisiae*)

UFFE H. MORTENSEN[†], CHRISTIAN BENDIXEN[‡], IVANA ŠUNJEVARIC[§], AND RODNEY ROTHSTEIN[¶]

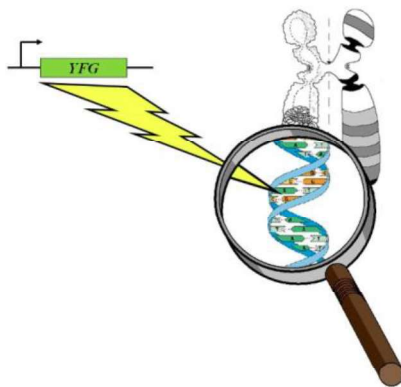
Department of Genetics and Development, College of Physicians and Surgeons, Columbia University, 701 West 168th Street, New York, NY 10032

Communicated by Paul Berg, Stanford University Medical Center, Stanford, CA, July 3, 1996 (received for review May 8, 1996)



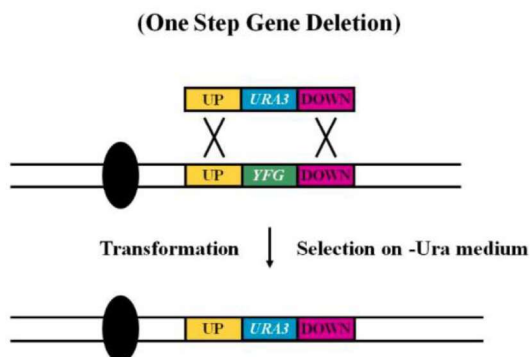
• Lecture 5: Recombination in Biotechnology

Gene Targeting:

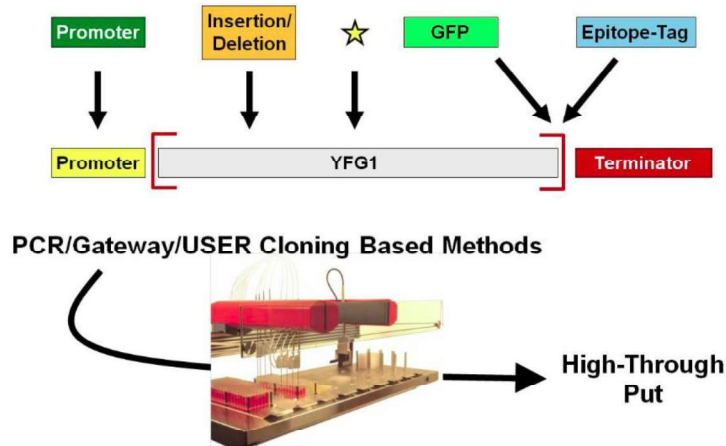


The ability to tailor a piece of DNA in a test tube and then take that piece of dna and put it into a cell. A circle floating around outside of a chromosome that piece of dna can go into a chromosome in between a YFG gene specifically.

One step gene deletion:

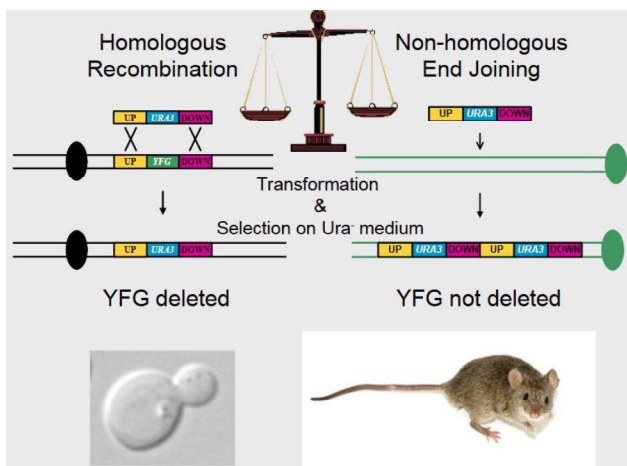


The reason it is linear is DNA double-strand break repair. In yeast this is super efficient. I do not have the time to do this, but you can make all kind of varieties to this theme, you can say that you for example do not like a promoter and you can replace the promoter.

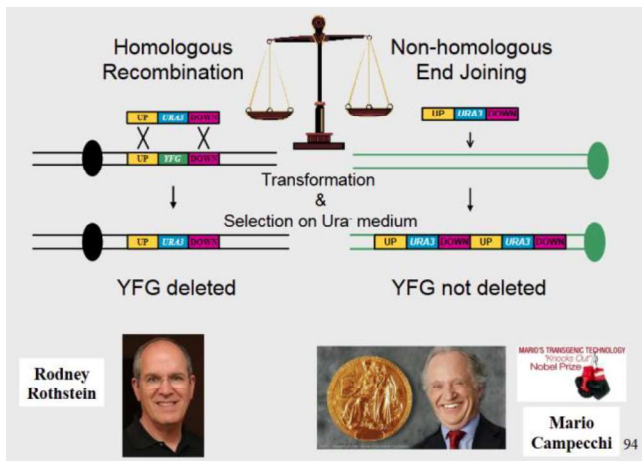


The number of experiments done by this theory is uncountable.

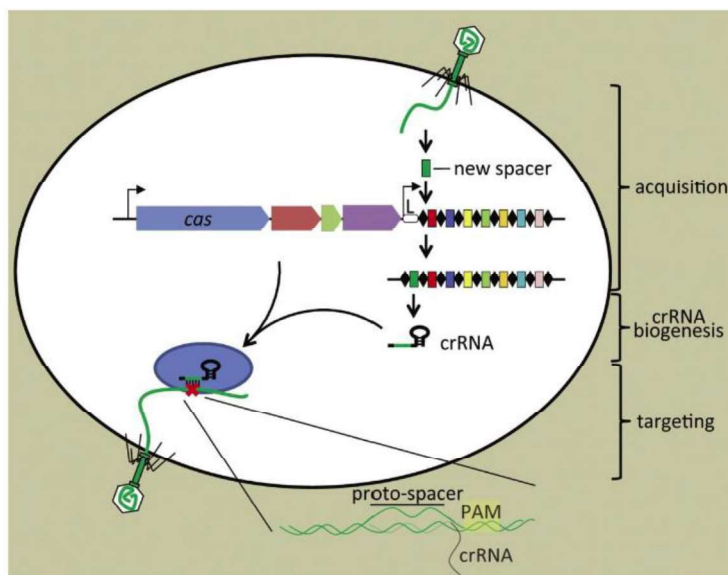
But there is a problem. It is hard to make a DNA double-strand break.



Basically only in bakers yeast, but in the mouse, difficult.

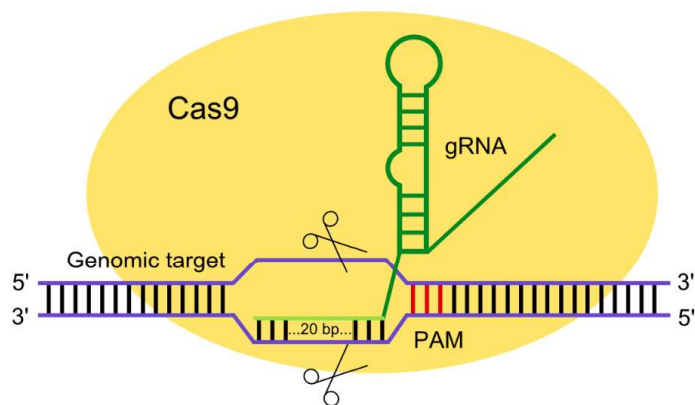


DNA Nucleases may defend against invading DNA: CRISPR/Cas9 - Sequence-Specific Adaptive Immunity:



CRISPR: Clustered Regularly Interspaced Short Palindromic Repeats

The e-coli has a RNA nuclease which cleaves DNA, so if you redesign/recombine this RNA nuclease and use gene targeting by telling it where it is supposed to cleave you can make the dna souble strand break.



When the CRISPR sequence sits at the chosen site it can make a DNA double-strand break.

Specific Ploidy Ablation – SPA:

Aim of Technology: High Throughput Transfer of a Plasmid from a Donor Baker's Yeast Strain to an arrayed recipient Yeast Library.



Example: The Plasmid encodes Your Favorite Protein tagged with GFP

Question: Which Mutations in the Library influences its Position in the Cell?

Molecular Biology Week 11:

Learning objectives

Notes from the lecture

Molecular Biology Week 12:

Learning objectives

Notes from the lecture

Molecular Biology Week 13: Recap and test exam

Learning objectives

Notes from the lecture

Test exam

Study number:

27026 Molecular Biology

Written exam, December 2024,

Duration: 4 hours

All aid is allowed except personal communication and internet access.

Answers must be written in the boxes in the word document, and not exceed the provided space. Remember to write exam number at the top. Upload your answers as a pdf file.

The exam is divided into three parts, A, B and C. Part A is 20 Multiple Choice questions; part B is 6 short questions, while part C is one large problem with associated questions. A, B and C count for 30 %, 35 % and 35 % of the final grade.

The Multiple-Choice part is accessed separately in the Multiple-Choice Module at DE. Each question counts 1.5 % of the total grade.

Part A. Multiple-Choice (30 % - 1.5 % for each question)

The Multiple-Choice part is accessed in the Multiple-Choice Module at DE.

	correct answer
1. Which of the following types of PCR is best suited for quantifying the amount of DNA in a sample?	
A) Reverse transcription PCR (RT-PCR), which is used to convert RNA into DNA before amplification. B) Colony PCR is a method to identify whether a DNA fragment is inserted into a plasmid without prior DNA purification. C) Nested PCR uses two sets of primers to increase specificity and reduce non-specific amplification. D) Quantitative PCR (qPCR) monitor DNA amplification during each cycle by measuring fluorescence. E) Conventional PCR is a method which allows a “target” DNA sequence to be selectively amplified several million folds in just a few hours.	Types of PCR
2. A type II DNA topoisomerase changes the linkage number of a supercoiled double-stranded DNA circle in steps of ...	
A) 0 B) 1 C) 2 D) 4 E) 8	Read into this
3. Occasionally uracil is found in DNA as the result of deamination of a specific base – which base?	
A) Adenine B) Cytosine C) Methylated Cytosine D) Guanine E) Thymine	Get a good picture of nucleotide structures
4. Many DNA polymerases contain a 3-5 exonuclease activity. This activity is mainly used to ...	
A) remove a misincorporated nucleotide during DNA synthesis B) degrade the RNA primer after DNA synthesis C) degrade viral DNA during DNA replication D) degrade telomeres during DNA synthesis E) degrade Okazaki fragments after DNA synthesis	Look into the different DNA-polymerases Read up on telomeres and telomerase
5. DNA replication requires helicase activity. This enzyme catalyzes	
A) degradation of RNA primers B) synthesis of the RNA primer C) removal of superhelical stress	Look into helicase

D) DNA synthesis E) separation of the two strands in double-stranded DNA	Opens up the helical coils to read the dna (correct?)
6. Crossing-over events are the result of the resolution of Holliday junctions. How many Holliday junctions are involved in formation of a single crossing-over event	
A) 1 B) 2 C) 3 D) 4 E) 8	What is a holliday junction
7. The DNA repair pathway Non-homologous End-joining is used to ...	
A) seal DNA nicks without using a template B) seal DNA nick using the sister chromatid as template C) seal DNA nicks using a homologous chromosome as template D) seal a DNA double-stranded break without using a template E) seal a DNA double-stranded break using the sister chromatid as template	Understand what this and the consequences of it “non-homologal = without a template”
8. In the genome, a DNA transposon is flanked by a small direct repeat of host DNA. This is because ...	
A) the transposase cleaves the target site to produce a DNA double strand break with staggered ends prior to integration of the transposon B) the transposase cleaves the target site to produce a DNA double strand break with blunt ends prior to integration of the transposon C) the transposase cleaves only target sites with a direct repeat prior to integration of the transposon D) the transposase duplicates the host ends at the old-integration site prior to integration into a new site E) the transposase possesses a non-template-based polymerase activity	
9. Which of the following best describes a key difference between bacterial and eukaryotic transcription initiation?	
A) The bacterial RNA polymerase holoenzyme directly recognizes the promoter sequences without the need for additional factors, while in eukaryotes, RNA polymerase II requires the assembly of a preinitiation complex with multiple general transcription factors at the promoter. B) In eukaryotes, transcription initiation is driven primarily by sigma factors binding directly to promoter elements, while bacterial transcription initiation relies on the TATA-binding protein bound to the RNA polymerase core enzyme.	(sigma factors are only for prokaryotes) Holoenzyme is RNA polymerase +

C) Bacterial transcription initiation occurs at TATA boxes, while eukaryotic transcription initiation occurs only at GC-rich regions. D) In bacteria, RNA polymerase initiates transcription at regions rich in enhancer sequences, whereas eukaryotes do not require enhancer sequences for transcription initiation. E) Bacterial transcription initiation requires RNA polymerase II, while eukaryotic transcription initiation requires RNA polymerase I.	the sigma factor TATA-box is only for eukaryotes
10. Which of the following best describes the role of TF _{II} D in eukaryotic transcription initiation?	
A) TF_{II}D phosphorylates the RNA polymerase II CTD to activate it for transcription elongation. B) TF_{II}D binds directly to enhancer sequences to facilitate RNA polymerase II recruitment. C) TF_{II}D recognizes and binds to the TATA box within the promoter, serving as an anchor for the assembly of other general transcription factors and RNA polymerase II. D) TF_{II}D stabilizes the RNA-DNA hybrid during transcription to prevent premature termination. E) TF_{II}D directly catalyzes the addition of ribonucleotides to the growing RNA strand	Read up on TF_{II}D If no TATA-box TF_{II}D bind to the IMR (what is this)
11. Which of the following processes is NOT involved in the maturation of eukaryotic pre-mRNA into mature mRNA?	
A) Addition of a 5' cap B) Splicing C) Polyadenylation at the 3' end D) Editing of the mRNA by removing the 5' cap and replacing it with a Shine-Dalgarno sequence. E) Alternative splicing, allowing the production of multiple protein isoforms from a single gene.	Shine-dalgarno sequence happen I prokaryotes
12. Which of the following statements best describes a common mechanism of mRNA degradation in bacteria?	
A) Bacterial mRNA is degraded primarily by the exosome complex, which recognizes poly(A) tails. B) mRNA degradation in bacteria often begins with the removal of the 5' cap, followed by exonuclease activity. C) mRNA degradation in bacteria typically involves endonuclease cleavage, followed by exonuclease activity that rapidly degrades the RNA from the cleavage site. D) Bacterial mRNA is highly stable due to extensive 3' polyadenylation, preventing rapid degradation. E) mRNA degradation in bacteria is regulated by specialized spliceosomal components that target unspliced RNA for breakdown.	Read up on general eykaryote vs prokaryote
13. Which of the following statements describes best the roles of initiation factors in bacterial translation initiation?	

A) IF-1 binds to the A site of the small ribosomal subunit to prevent premature tRNA binding, while IF-2, in its GTP-bound form, facilitates the recruitment of the initiator fMet-tRNA to the P site. IF-3 binds to the E site to prevent premature association of the ribosomal subunits. B) IF-2 binds directly to the Shine-Dalgarno sequence to position the ribosome at the start codon, while IF-1 and IF-3 recruit the large ribosomal subunit. C) IF-3 recognizes and binds to the Shine-Dalgarno sequence, and IF-1 catalyzes GTP hydrolysis to allow fMet-tRNA to enter the P site. D) IF-1 binds to the Shine-Dalgarno sequence to stabilize ribosome-mRNA interactions, IF-2 blocks premature tRNA binding, and IF-3 promotes ribosomal subunit joining. E) IF-3 removes any incorrectly bound tRNAs from the small ribosomal subunit, while IF-2 recruits elongation factors to facilitate rapid translation once the start codon is recognized.	Look up on the roles of initiation factors in bacterial translation initiation
14. In the context of prokaryotic gene expression regulation, which of the following statements regarding repressors is FALSE	
A) a loss of function mutation in a gene encoding for a repressor results in high expression of the structural gene controlled by the repressor B) a loss of function mutation in a gene encoding a repressor results in low expression of the structural gene controlled by the repressor C) repressors bind to the operator sites overlapping the promoter of the controlled structural gene(s) D) some repressors need binding of a co-repressor to adopt the conformation that enables binding to its operator site E) a repressor is a trans-regulatory element	What is a trans-regulatory element
15. In the context of prokaryotic gene expression, a promoter is...	
A) a cis-regulatory element B) a trans-regulatory element C) a dimeric protein D) a palindromic sequence E) a DNA binding domain	Look up on all of this
16. If boundary elements (insulators) would not exist, what would happen if histone acetyltransferases (HAT) start relaxing chromatin around histones?	
A) HAT will eventually stop acetylating histones when it finds a specific DNA sequence B) HAT are sequence-independent and therefore will relax all the chromatin around the histones C) boundary elements (insulators) are not present in eukaryotes D) HAT will eventually recruit a Histone deacetylase (HDAC) to stop the relaxation of the chromatin by removing acetyl groups from the histones E) HAT will eventually release its ATPase domain stopping its activity	Look into HAT They need something telling them where to go (??)
17. In the context of eukaryotic gene expression regulation, which of the following statements regarding true activators is TRUE	

A) true activators always activate the expression of genes (positive control) B) true activators are always inside the cell nucleus C) true activators are heterodimers that make direct contact with the basal apparatus through a co-activator D) true activators can have a DNA binding domain and an activator domain (that makes contact with the basal apparatus) E) true activators recruit chromatin-remodelling complexes	Read up on true activators They can be both positive and negative control.
18. Why does ribosome-mediated attenuation mechanism only happen in prokaryotes?	
A) Because eukaryotic genes do not have leader sequences B) Because prokaryotes have more ribosomes than eukaryotes C) Because in prokaryotes transcription and translation processes are coupled. D) Because in eukaryotic leader sequences are not translated E) Because in prokaryotes the metabolite binding inducing the change in the leader sequence are readily available while in eukaryotes metabolites cannot enter the nucleus	Read up on this: ribosome-mediated attenuation mechanism
19. Which of the following types of RNA can regulate multiple targets due to an imperfect match with the target nucleotide sequence	
A) siRNA B) tRNA C) riboswitches D) mRNA E) miRNA and asRNA (micro RNA(in eukaryotes) and antisense RNA(cis-encoded have perfect match for trans encoded and they match with more multiple targets)	Read up on the different types of RNA and riboswitches
20. Regarding the use of viruses as biotechnological tools, which of the following statements regarding adenoviruses is FALSE .	
A) adenoviruses integrate their genetic material into the host cell genome B) adenoviruses have broad cell tropism C) adenoviruses provide transient immune expression D) adenoviruses have high transduction efficiency E) adenoviruses have a high capacity for carrying foreign DNA	What are adenoviruses?

Part B. Short questions (35 %).

Problem B1. (6 %).

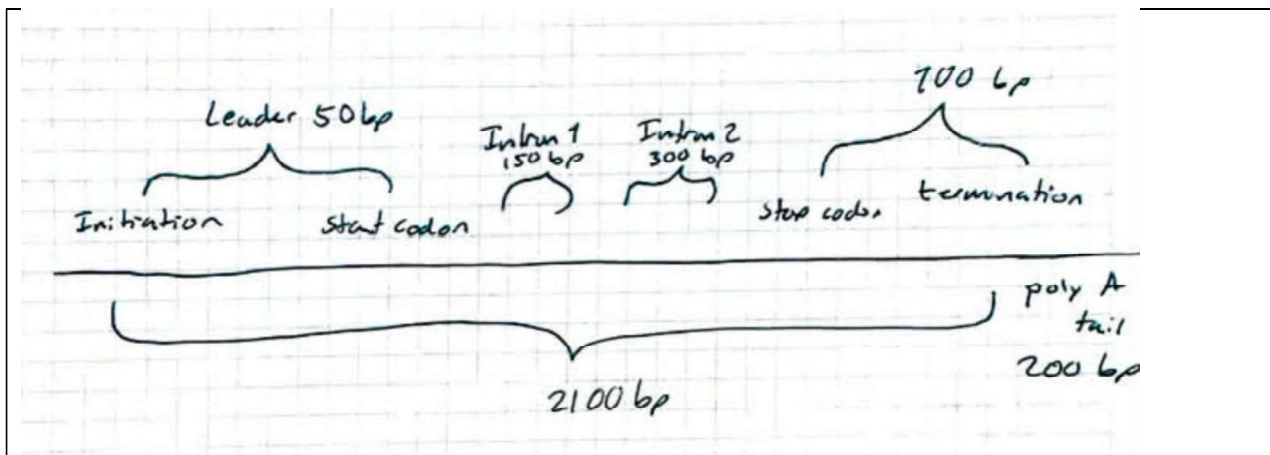
Read up on northern blot

In the figure below, the result of a Northern blot experiment is shown. At time $t = 0$ minutes Rifampicin was added to growing cultures of <i>E. coli</i> wildtype (wt) and a derived strain with a loss-of-function mutation in the <i>rne</i> gene (<i>rne-537</i>) encoding RNase E. Samples were taken at the specified timepoints, and RNA was extracted and analysed by a Northern blot. The probe was DNA complementary to the small RNA CsrB.	
The figure shows a Northern blot for CsrB mRNA. The wildtype (wt) strain shows a rapid decrease in mRNA levels over time, while the <i>rne-537</i> mutant strain shows a much slower rate of degradation.	
Give a rough estimate of the half-life of the <i>csrB</i> mRNA in the wildtype and <i>rne-537</i> mutant <i>rne-537</i>	For wildtype: 2min (less than 5 min is ok) For mutant: 20 min (more than 5 min is ok)
How can you explain the change in half life in the <i>rne-537</i> mutant?	RNase – an enzyme that degrades RNA, all RNA is going to have an extended half-life as the RNase has loss-of-function -so the half-life is now longer. Ezperanzas: “RNase E is the initial step in mRNA degradation in <i>E. coli</i>. When RNaseE is missing RNA in general is more stable.”

Problem B2. (5 %).

A eukaryotic gene with two introns of 150 and 300 bases, and the distance from transcription initiation to the terminator is 2100 bases. The leader is 50 bases and the distance from the stop codon to the transcription terminator is 100 bases. If you assume that a polyA tail is 200 bases, calculate a) the length of the mature mRNA; and b) the approximate number of amino acids in the protein encoded from the gene.

Drawing to visualize:



Length of mature mRNA: introns bliver splejset ud og polyA-tail bliver added så: $2100 - 450 + 200 = 1850$

Length of protein: $2100 - 50(\text{leader}) - 100(\text{between stop codon and termination}) - 150(\text{intron 1}) - 300(\text{intron 2}) = 1500$ bp, for protein we need it in amino acids so $1500/3 = 500$ amino acids.

Problem B3. (6 %).

The *Escherichia coli* *ibsD* gene encodes a polypeptide belonging to a class of very small toxic bacterial polypeptides. The DNA sequence of the coding strand is shown below, with the 48 amino-acid open reading frame indicated by underscoring. To clone the *ibsD* along with necessary translation initiation elements in an expression vector, PCR amplification of the region was performed. In the scheme below, propose two 20-base primers that would yield the desired product and provide the sequence of the first five amino acids of the IbsD peptide.

Only using canonical sequences even though real life would look differently – so only look for canonical sequences given in slides

20 40 60
5' -CTCCCAAGAAGGGGCCTTACAGAAGAGGAGGTTATGATGAAGCTCGTCATCATACTGATTGTGTTGTT-3'
80 100 120 140
5' -ACTCGTAAGTTTCGCAGCTTATTAACATGCCAATCAGAGGGGGGAGAAATCCTCCCTCACCTTGTTTCCTT-3'

Table 2.5 The genetic code as expressed by triplet base sequences of mRNA^a

Codon	Amino acid	Codon	Amino acid	Codon	Amino acid	Codon	Amino acid
UUU	Phenylalanine	UCU	Serine	UAU	Tyrosine	UGU	Cysteine
UUC	Phenylalanine	UCC	Serine	UAC	Tyrosine	UGC	Cysteine
UUA	Leucine	UCA	Serine	UAA	None (stop signal)	UGA	None (stop signal)
UUG	Leucine	UCG	Serine	UAG	None (stop signal)	UGG	Tryptophan
CUU	Leucine	CCU	Proline	CAU	Histidine	CGU	Arginine
CUC	Leucine	CCC	Proline	CAC	Histidine	CGC	Arginine
CUA	Leucine	CCA	Proline	CAA	Glutamine	CGA	Arginine
CUG	Leucine	CCG	Proline	CAG	Glutamine	CGG	Arginine
AUU	Isoleucine	ACU	Threonine	AAU	Asparagine	AGU	Serine
AUC	Isoleucine	ACC	Threonine	AAC	Asparagine	AGC	Serine
AUA	Isoleucine	ACA	Threonine	AAA	Lysine	AGA	Arginine
AUG (start) ^b	Methionine	ACG	Threonine	AAG	Lysine	AGG	Arginine
GUU	Valine	GCU	Alanine	GAU	Aspartic acid	GGU	Glycine
GUC	Valine	GCC	Alanine	GAC	Aspartic acid	GGC	Glycine
GUA	Valine	GCA	Alanine	GAA	Glutamic acid	GGA	Glycine
GUG	Valine	GCG	Alanine	GAG	Glutamic acid	GGG	Glycine

Primers for PCR. Give the sequence of the primers in the 5'-3' direction

You always have to put the forward primer in 5'-3' direction, for the reverse remember to do the reverse complement.

Answer:

Forward; 5'-GAAGGGGGCCTTACAGAAGA-3'

Reverse: 5'-GGATTCTCCCCCTCTAGA-3'

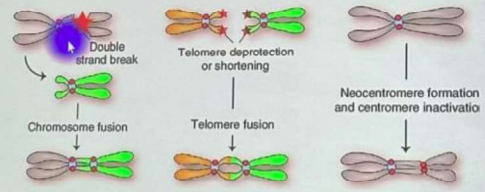
Sequence of the first five amino acids

Find start codon – ATG(AUG in RNA) (the first methionine and read 3 bp at a time and translate to aminoacids): Met-Met-Lys-Leu-Val

At position 96, a potential AUG start codon is found. Why does it not result in translation initiation?

Answer: No Shine-Dalgarno sequence in front.

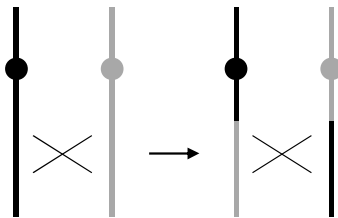
Problem B4. (6 %).

Telomeres protect the ends of the linear eukaryotic chromosomes. As human cells divide, telomeres are getting shorter. Explain why.	Problem B4 (6%) Aging human cells may not have telomeres and such ends will be a substrate for the non-homologous end-joining DNA repair pathway. What problem can that cause? Describe one possibility. The linear ends are no longer protected from DNA repair pathways like non-homologous end-joining. As a result, chromosomes may fuse or circularize. In particular, the former is not good as di-centric chromosomes are formed. Alternatively, shortening of chromosome ends will proceed into coding regions and this will cause gene loss.  Answer: All DNA polymerases need a primer. In this case an RNA primer. As a result, the 5'-3' end of a chromosome cannot be fully replicated and will shorten after each division. As telomerase is not active except in the zygote and early stem cells, chromosomes will shorten by a little bit in each round of division.
Aging human cells may not have telomeres and such ends will be a substrate for the non-homologous end-joining DNA repair pathway. What problem can that cause? Describe one possibility.	Answer: The linear ends are no longer protected from DNA repair pathways like non-homologous end-joining. As a result, chromosomes may fuse or circularize. In particular, the former is not good as di-centric chromosomes are formed. Alternatively, shortening of chromosome ends will proceed into coding regions and this will cause gene loss.
Now that we know that loss of telomeres is associated with ageing; what could be the advantage of not having active telomerase in fully developed cells?	Answer: Cells with telomeres will have endless division potential. Sounds good, but this is what characterizes tumor cells. Indeed, telomerase is often active in tumor cells.

Problem B5. (6 %).

The Figure below shows the result of a crossover between two linear homologous	Answer: The genetic diversity of the gametes are vastly increased and this will speed up
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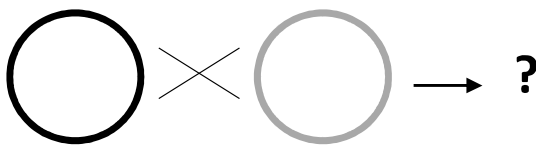
chromosomes during meiosis. Such crossovers are very frequent. Explain briefly why it is a good idea to generate chromosomes that are a mix of the two parental homologs.



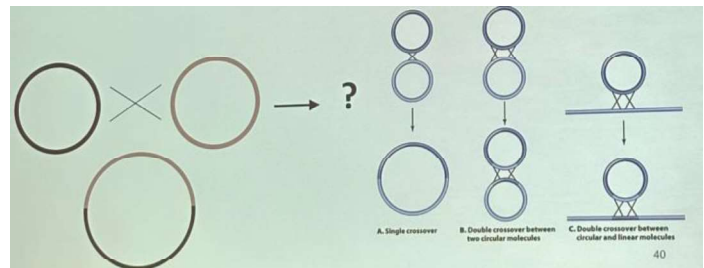
evolution to ensure survival of the species

Consider two circular homologous chromosomes that are performing a single homologous crossover event. See drawing below.

Make a drawing that shows the outcome of the crossover event.



A chromosome that includes both of them – it has twice the size



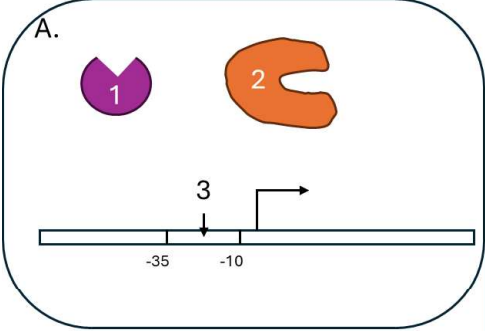
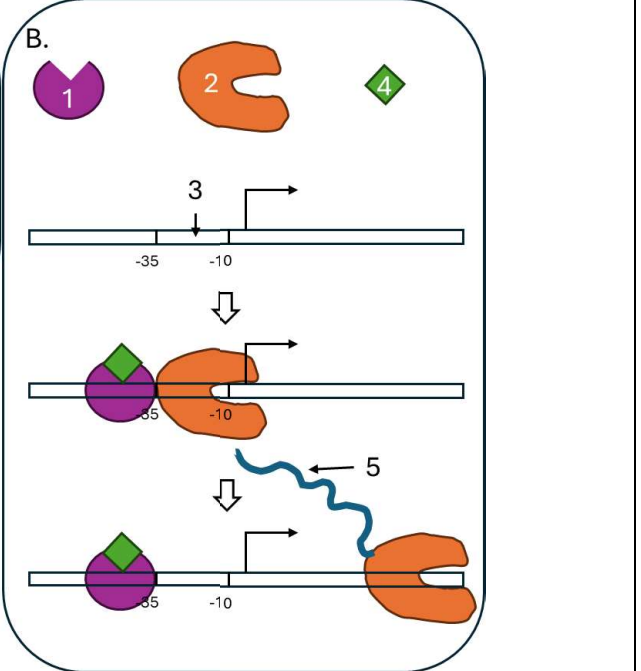
Based on the result of the crossover between circular chromosomes, provide an explanation for why eukaryotes use linear chromosomes.

To keep the size and the structure of the information – otherwise the chromosomes are not gonna look the same, resulting in genetic instability (?)

Answer:

The outcome of recombination between two circles will often yield a single fused circle rather than two circles. Hence, a meiotic division with crossing over as we know it from eukaryotes will most often produces gametes with chromosomes that are not similar to the parental chromosomes. Organisms with a circular chromosome can therefore not evolve via a meiotic style sexual cycle

Problem B6. (6 %). Respond to the following questions regarding the attached figure that illustrates a regulation of gene transcription situation

A. 	B. 
Does the figure describe regulation in a bacterium or a eukaryote. Briefly state why.	It is a Prokaryote – because the promoter is showing -10 and -35 and that is a prokaryote promoter
Name the molecules/structures numbered in the figure	1. Activator
	2. RNA polymerase
	3. Promotor
	4. Inducer
	5. mRNA
What kind of regulation is illustrated in the figure	Inducible activation
What would be the consequence regarding gene transcription if a mutation on “1” would make it incapable of binding “4”	It would cause a loss-of-function - The activator (1) will be hold in the inactive conformation in the presence or the absence of the coactivator (4) therefore the expression of the controlled gene would be low

Part C. (35 %).

The **eukaryotic gene CEN1** is crucial for cellular growth regulation in response to the growth hormone **EGF** (Epidermal Growth Factor)

Overexpression of **CEN1** is associated with abnormal tissue growth and has been implicated in various cancers, such as hepatocellular carcinoma and breast cancer.

Question 1 (5 %). Describe in chronological orders the steps necessary to express a eukaryotic protein (transcription and translation included). Do not explain in detail the events in the individual steps – use headings only.

Chromatin opens is veryyy important to mention!!!

Answer:

1. Chromatin opens (by the action of histone remodelling enzymes)
 2. Transcription factors are bound (to enhancer regions by direct contact with the DNA
 3. Transcription-activator domains recruit the transcription apparatus by direct interactions with their elements)
 4. The RNA pol II transcribes a gene (into pre-mRNA)
 5. mRNA is processed (by splicing, capping and polyadenylation)
 6. The mature mRNA is translocated to the cytoplasm
 7. mRNA is translated into protein (and mRNA is degraded)
- (The answers are also correct without the text in bracket

To investigate the transcriptional regulation of CEN1, researchers developed a reporter cell line where the promoter of CEN1 is fused to a luciferase reporter gene. The luciferase activity (in relative light units, RLU) was measured in the presence or the absence of the hormone EGF in wild-type cells and in three cell lines with three unknown mutations (= Mutant 1-3) - see **Table 1**.

Table1

	Control (wild type)		Mutant 1		Mutant 2		Mutant 3	
EGF (hormone) (+ present / - absent)	-	+	-	+	-	+	-	+
Luciferase activity (relative	20	100	20	20	100	100	5	25

light units – RLU)								
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The CEN1 gene is believed to be regulated by two transcription factors, **TF-A** and **TF-B**, which may act as either activators or repressors depending on the context. To investigate which gene is mutated in each of Mutant, 1,2 and 3, the cell lines were transformed with a plasmid encoding a functional copy of TF-A or TF-B (**pTF-A** and **pTF-B**), and subsequently the luciferase activity was determined following grown in the absence or presence of EGF (see **Table 2**). In the wild-type control, introduction of the pTF-A or pTF-B plasmids did not lead to changes in the expression of the reporter gene – see left column in Table 2.

Table 2.

	Control (wild type)		Mutant 1				Mutant 2				Mutant 3			
Transformed plasmid			+pTF-A		+pTF-B		+pTF-A		+pTF-B		+pTF-A		+pTF-B	
EGF (hormone) (+ present / - absent)	-	+	-	+	-	+	-	+	-	+	-	+	-	+
Luciferase activity (relative light units – RLU)	20	100	20	100	20	20	100	100	20	100	5	25	5	25

Question 2 - (5 %). Indicate what gene that might be mutated in Mutant 1, 2, and 3 and state if the mutated element is a cis-regulatory element or a trans-regulatory element.

Mutant 1: **TF-A-trans** (because it can get substituted and restored when substituted with TF-A)

Mutant 2: **TF-B-trans** (because it can get substituted and restored when substituted with TF-B)

Mutant 3: **promotor-cis** (how ofte the RNA polymerase is recruited to a site, so when the numbers genereally drop it is a hint that it is a promotor – cis because it cannot be substituted)

Ezperanza Answer:

Mutant 1 is mutated in TF-A. It is a trans-regulatory element

Mutant 2 is mutated in the TF-B. It is a trans-regulatory element

Mutant 3: it is mutated in the promotor. It is a cis-regulatory element

Question 3 – (5 %). Assign a function to TF-A and TF-B in control of CEN1 expression and indicate the role of EGF in the regulation by TF-A/TF-B

TF-A is a activator of CEN1 in the presence of EGF

TF-B is an repressor of CEN1 in the absence of EGF.

To further investigate how TF-A may regulate gene expression, the researchers searched for functional domains in the annotated TF-A sequence. They succeeded in finding a putative homodimerization domain, and a domain that can make contact with the basal apparatus. However, the researchers were unable to identify a DNA binding domain.

Question 4 - (5 %) Explain how is it possible that TF-A controls the expression of CEN1 without a DNA-binding domain

Transcription factors have a modular nature – TF-A can interact with an independent DNA-Binding domain directly or via a co-activator - as long as it is in contact with something that has a DNA-binding-domain it is possible to control expression of CEN1.
--

Researchers studied miRNA expression in a healthy cell line and a cancer cell line and identified a specific miRNA, miR-001, whose expression is significantly altered in the cancer cell line. During *in vitro* experiments, they observed that native miR-001 forms a stable secondary structure which may abolish its function. To study the function of miR-001 in cell proliferation, the researchers generated a miR-001* mutant with reduced tendencies to fold into the predicted secondary structures and transfected this mutation into the cancer cell line. They then measured the following parameters: miR-001RNA levels of, and Argonaute:miR-001 complexes; protein levels of EGF and CEN1; and the overall growth rate of the three cell lines (**Table 3**).

Table 3

Cell line	miR-001 (pM)	Argonature:miR-001 (pM)	EGF (pM)	CEN1 (pM)	Growth rate (relative units)
Control (wild type)	10	100	5	10	100
Cancer cell line	0.01	0.1	50	100	300
Cancer cell line + miR-001* mutant	10	100	50	10	200

Question 5 - (3%). Name two methods the researchers could use to measure RNA levels of pre-miR-001, and mature miR-001.

Answer:

Reverse transcriptase PCR followed by qPCR of the cDNA, (qRT-PCR)

Northern Blotting.

Question 6 – (6 %).

How does the level of miR-001 correlate with the concentrations CEN1 in the control and cancer cell lines?

Argonature: miR-001 (pM) high - growth rate of CEN1 is low
Argonature: miR-001 (pM) low - growth rate of CEN1 is high

Ezperanza Answer:

High miR-001 levels in the wt correlate with low CEN1 concentration. In the cancer cell line drastically reduced miR-001 levels, result in CEN1 concentration being significantly elevated.

Describe how the growth rate is correlated to CEN1 expression in the control and cancer cell lines

High Cen1 correlates with higher growth rate

Low Cen1 correlates with lower growth rate

Ezperanza Answer:

In the control cell line, CEN1 expression is helping to maintain controlled cell growth. In contrast, the cancer cell line shows

	drastically high CEN1 concentration, and the growth rate is much higher (300 units)
What is the role of EGF in regulating CEN1 expression and growth rate in the cancer cell line?	Ezperanza Answer: In the cancer cell line EGF levels are higher, which likely stimulate CEN1 expression through an activator factor or removing a repressor (which can be TF-A and TF-B).

Question 7 – (3 %). Describe in a few sentences the role of Argonaute proteins in gene regulation.
What is argonaute, what is it doing and why is it important: When the microRNA or a silencing RNA argonaute will recruit something to decrease translation (right?) Ezperanza answer: Argonaute proteins bind to small RNAs, such as miRNAs or siRNAs forming the RISC complex. Argonaute proteins guide RISC to specific mRNAs based on the sequence complementarity between the small RNA and the mRNA. The binding of the RISC complex prevents translation of the mRNA

Question 8 – (3%) Explain how miR-001* mutation affects the concentrations of EGF and CEN1, and describe its impact on the growth rate of the cancer cell line.
Ezperanza answer: miR-001 Mutant 1 restores miR-001 levels to those found in healthy cells by increasing the formation of Argonaute:miR-001 complexes to normal levels. This leads to a significant reduction in CEN1 concentration, bringing it close to normal levels and mitigating its cancer-associated overexpression. However, EGF levels remain elevated, indicating that while CEN1 expression is controlled, EGF is not fully repressed. As a result, the growth rate of the cancer cell line decreases compared to the untreated cancer cell line but remains higher than in the control (healthy) cells

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Part A. Multiple Choice (40 % - 2 % for each question)

1.

What is true about bacterial reporter gene fusions?

In translational fusions, the reporter gene is expressed from its own Shine-Delgarno sequence.

In transcriptional fusions, the start codon of the reporter gene is deleted.

The reporter gene should be easy to assay. **x**

The *lacZ* gene from *E. coli* cannot be used as a reporter gene in yeast.

2.

qPCR is used to

quantify RNA.

map the 5' end of a mRNA.

map the 3' end of a mRNA.

quantify DNA. **x**

3.

Which of the following is a telomeric function?

Supercoiling of DNA

Packaging a phage genome into its capsid

Protection of the chromosome end **x**

Providing a site of origin for DNA replication

4.

Initiation of DNA replication of a eukaryotic chromosome requires:

an RNA Primer **x**

a single-stranded DNA primer

a double stranded DNA primer

a tRNA

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5.

DNA polymerases often possess an exonuclease activity, which is used to:
degrade undesired DNA invading a cell (e.g. a phage)
remove a misincorporated nucleotide. **x**
speed up DNA synthesis
separate strands prior to DNA replication

6.

One possible consequence of branch migration of a Holliday junction is:
Resolution of the Holliday junction.
Formation of DNA single stranded breaks.
Recruitment of Spo11.
formation of heteroduplex DNA. **x**

7.

Which genetic element is obligatory for a plasmid to transfer by conjugation?

An inverted repeats (IR).

An antibiotic marker.

Flagella.

An *oriT* **x**

8.

Which statement regarding eukaryotic transcription initiation is **TRUE**?

The TATA box Binding Protein (TBP) is only required for transcription initiation at RNA pol II promoters.

RNA pol II promoters can be identified by the presence of a TATA box.

The expression of all types of promoters is dependent on chromatin structure. **x**

RNA polymerase III transcribe 18s/28s rRNA.

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9.

The transcription initiation of eukaryotic RNA polymerase II promoters is characterized by sequential events.

- A. RNA polymerase II binding.
- B. Phosphorylation of the CTD domain of the RNA polymerase.
- C. Binding of Transcription Factor TF_{II}D.
- D. Promoter clearance.

Which order of events is correct?

- 1. C – A – B – D
- 2. A – C – B – D
- 3. C – A – D – B
- 4. A – C – D – B

10.

In intron excision, two transesterification steps occur. At which complex formation step in the process does it happen?

E complex.

A complex.

B complex (B1 and B2).

C complex (C1 and C2). **x**

11.

Which statement is **NOT** true for alternative splicing of pre-mRNAs?

It can change the target compartment for the protein encoded by the mature mRNA.

Alternative splicing can be a way to change the phenotype of a cell (e.g. change of sex).

Alternative splicing requires at least 2 introns in the same pre-mRNA. **x**

There can be more than one splice site in a single intron.

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12.

What is the function of the A-site in the ribosome?

It binds the fMet-tRNA in the initiation of the translation.

It binds the tRNAs before the amino acid is transferred to the peptide being synthesized. **x**

It is the site related to release of the finished peptide after translation is complete.

It is the site that holds the growing peptide during translation, until it is released from the ribosome.

13.

If the expression of a prokaryotic gene is subject to regulation by an activator, the following statement is **TRUE**?

A loss of function mutation in the activator gene leads to constitutive high expression.

If a co-activator is required, the gene expression is low in the presence of the co-activator.

A mutation in the activator gene that means that the activator is trapped in its activating conformation, will result in constitutive high expression. **x**

Activators compete with RNA polymerase for DNA binding.

14.

Which statement about activators is **FALSE**?

Activators may work by making protein-protein contact with basal transcription factors.

Activators may work by covalent histone modification.

Activators determine the frequency of transcription initiation.

Activators are often facilitating heterochromatin formation. **x**

15.

Which statement about the chromatin remodelling is **TRUE**?

Histone acetylation facilitate heterochromatin formation.

Chromatin remodelling occurs only during DNA replication.

The Chromatin remodelling complex binds to the TATA box during transcription initiation.

Histone deacetylation is associated with repressor activity. **x**

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16.

siRNA is the same as

tRNA.

Antisense RNA.

Dicer.

Small interfering RNA. **x**

17.

Which of the following is **NOT** a mechanism in gene silencing by miRNA?

Competition for a cap structure.

Inhibition of ribosomal subunit joining.

De-adenylation and de-capping.

Enhancer degradation. **x**

18.

The T4 phage protects its DNA from host restriction endonucleases by:

integrating the viral genome into the host genome where it will not be degraded.

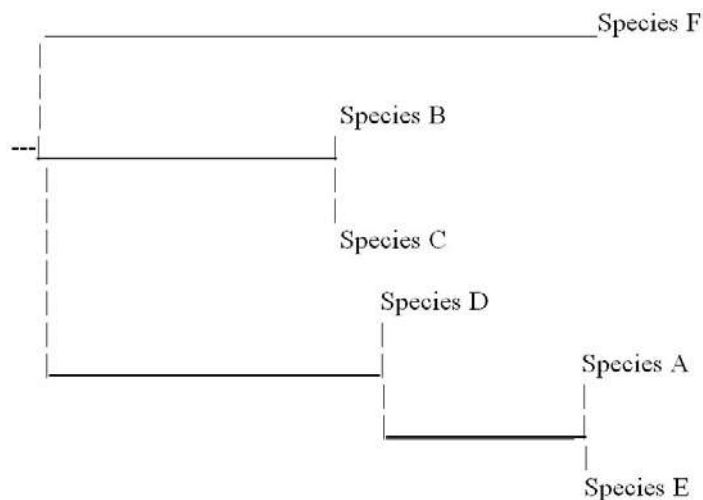
glucosylating cytosine bases in the T4 genome to prevent DNA cleavage. **x**

circularizing the viral genome so that it will not be degraded.

methylating all four bases (A, T, C, G) in the T4 genome to prevent DNA cleavage.

19.

Based on the phylogenetic tree below, which of the following statements is **TRUE**?



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Species F is more closely related to species A than to species B.
Species B is more closely related to species C than to species F. **x**
Species D is more closely related to species F than to species E.
Species F is more closely related to species E than to species D.

20.

The observation that, when several nucleotide triplets encode the same amino acid, one tends to be used in genes more frequently is called:

mutational bias.

genetic drift.

codon bias. **x**

synonymous substitution.

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Set B

1. (8 %). Genomes.

Explain the terms "core genome" and "pan genome" and describe how each contributes to the genome of a bacterial species. Give an example of genes that are part of a core genome versus those that are more often in the pan genome.

A core genome is the genome content present in all strains of a bacterial species. These are typically fundamental metabolic genes that combine to help define the bacterial species as a whole. The pan genome is the genome content that is found in only some of the strains within the species. The additional genes in the pan genome are what distinguish strains from one another. Typical genes in the pan genome may be for accessory functions such as virulence, degradation of a variety of substrates, or symbiosis. Often the pan genome contains mobile elements and plasmids.

2. (8 %). DNA repair.

Briefly describe the mismatch repair pathway and how it contributes to gene conversion.

Answer. Heteroduplex DNA is formed during the recombinational repair of DNA lesions like DNA double strand breaks or collapsed replication forks. The heteroduplex DNA is a consequence of the damaged DNA strand invading an intact copy of the same sequence with the aim of using it as a template for repair. The template may be present on the sister chromatid or the homologous chromosome. If the sequence used for repair is on the homologous chromosomes, the sequences may not be identical and there will be mismatched nucleotides in the heteroduplex. Such mismatches will be substrate for the mismatch repair pathway that will replace one of the nucleotides of the mismatch for one that matches the base pairing potential of the other nucleotide, hence, potentially leading to gene conversion, where genetic information in a given locus is either lost or gained.

3. (8 %). Splicing.

Briefly describe the steps and co-factors involved in the formation of the commitment complex (E complex) in excision of introns from pre-mRNAs.

Answer: (The pre-mRNA has several important sites surrounding the introns. The introns are usually flanked by the sequences GU- and -AG which are the 5'- and 3'- splice sites. Furthermore, there is a branch site near the 3'-splice site. (this part is not needed for the answer, it is just to set the scene)

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For formation of the E-complex the U1 RNP and SR protein bind to the 5'-splice site and BBP bind the branch site. The U2 protein (actually U2AF) binds the 3'-splice site. Finally, the SR protein will form a branch between U1 and U2, forming the E-complex (fig. 19.10 in the book)

4. (8 %). Chromatin remodelling.

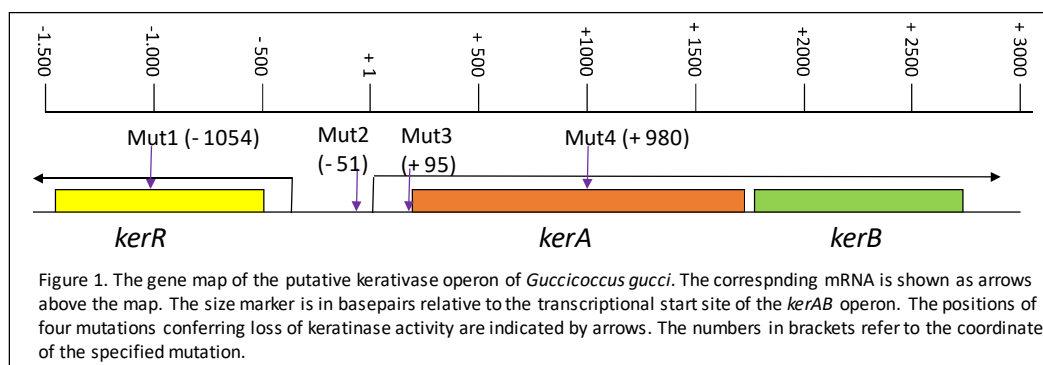
Chromatin remodelling is an active process in regulation of eukaryotic gene expression. Modification of histones is an important part of the mechanism. Briefly describe how histones are modified during the remodelling process. Include three common types of modifications.

Histones may be acetylated or deacetylated. Acetylation leads to less dense chromatin structure and thus increased expression. Alternatively, histones may become methylated. Methylation may result in both activation and de-activation of gene expression – dependent on the context. Histones may also be phosphorylated, and in this way change the chromatin structure.

Set C (28 % total)

The bacterium *Guccicoccus gucci* was isolated from the leather stock of an Italian producer of handbags. *G. gucci* was shown to be able to degrade keratin - one of the main constituents of skin. Keratin is a protein very rich in cysteine, and normally resistant towards proteases. *G. gucci* was found to produce two extracellular proteins: A proteolytic enzyme that hydrolyses peptides bonds, and disulfide bond-reducing enzyme that breaks the S-S bridges in proteins. None of these enzymes showed significant keratinolytic activity independently. However, after mixing of the two enzymes, the keratinolytic activity was increased more than 50-fold over that of the protease only. Interestingly, the genes encoding the enzymes were expressed significantly higher if *G. gucci* was grown in the presence of cysteine.

The genomic DNA from *G. gucci* was isolated and the genome sequence determined. A putative operon with homology to known keratinases was identified, and the map is shown in figure 1. The adjacent gene *kerR* showed homology to proteins with a Helix-turn-Helix motif.



To investigate the expression of the *ker* region, mutants unable to degrade keratin was investigated, and the transcription of the *kerAB* operon of both the wildtype and mutants were determined by Reverse Transcriptase PCR. The strains were grown in both the presence and absence of cysteine. The results of the Reverse Transcriptase PCR is shown in Table 1.

Table 1. Relative levels of *kerAB* mRNA as determined by Reverse Transcriptase PCR. The level of mRNA in the wildtype in the absence of cysteine was set to 1. The numbers in brackets refer to the coordinate of the mutation.

Strain/ supplement	Relative level of <i>kerAB</i> encoded mRNA				
	Wild type	Mut1 (- 1.054)	Mut2 (- 51)	Mut3 (+ 95)	Mut4 (+ 980)
No cysteine	1	1	1	1	1
With cysteine	200	1	1	200	200

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Question 1. (4 %). How could you confirm the transcriptional regulation of the *kerAB* operon using an alternative method to Reverse Transcriptase PCR?

Make a transcriptional fusion to a reporter gene, and measure the enzyme activity (eg β -gal)

Question 2. (4 %). The KerR protein showed homology to proteins with a Helix-turn-Helix motif. What does that signify?

A DNA binding protein

Question 3. (6 %). Based on the phenotype of the *kerR* mutant (Mut1) as shown in table 1, what kind of regulatory system is regulating *kerAB* expression?

KerR is most likely an activator.

Question 4. (4 %). What would you call cysteine in the *kerAB* regulatory system?

Co-activator

Question 5. (4 %). Considering that **no keratinase activity is observed** in Mut2, how would you explain its phenotype?

The activator-binding site is mutated

Question 6. (4 %). Considering that **no keratinase activity is observed** in Mut3, how would you explain its phenotype?

The Shine-Delgarno sequence is mutated – no translation of the protein, but the mRNA is still produced

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Question 7. (4 %). Considering that **no keratinase activity is observed** in Mut4, how would you explain its phenotype?

Loss-of-function mutation of the enzyme – no enzyme activity, but the genes are still transcribed and translated

Question 8. (4 %). Would you find a similar genetic organization of the *kerAB* genes in filamentous fungi? Argue for your statement.

No operons in eukaryotes as only the first start codon is used in translation

27026 Exam E2019

The exam set is divided in three parts, A, B and C. Part A is 20 Multiple Choice question; part B is 6 short questions, while part C is 1 large problem with 8 associated sub-questions. A, B and C counts a 40 %, 35 % and 25 % of the final grade.

Part A. Multiple Choice (40 % - 2 % for each question)

The Multiple Choice part is accessed in the multiple Choice Module of DTU Inside. Each question counts 2 % of the total grade.

Part B. Short question (35 %).

Question B.1. (6 %). Yeast replication

A common class of artificial yeast plasmids is of the CEN/ARS type. These plasmids have been constructed in the laboratory and are based on two native yeast sequences: an autonomously replicating sequence (ARS) and a centromere (CEN). Describe how the two features of the plasmid impact its copy number and stability. Include in your discussion, how many times the ARS sequence support replication initiation during S-phase, how many replicons that will appear during plasmid replication, and how the plasmid distributes between dividing cells.

Answer: The plasmid will behave (almost) like a chromosome. During S-phase it is replicated as a single replicon. There is only one ARS sequence. The plasmid will be replicated only one time due to its ARS sequence as eukaryotic origins of replication can only fire one time in each round of the cell cycle. During mitosis, the plasmid will be linked to the mitotic spindle via the centromere. This will ensure that each of the two resulting cells will receive one copy of the plasmid. For that reason, the copy number will be constant and the stability of the plasmid is high. (As a note, and not expected to be answered by student. The plasmids are not as stable as a real chromosome due to the small size and lack of telomeres)

Question B.2. (6 %). DNA repair

Three very common DNA lesions are: Firstly, base mismatches resulting from misincorporation by DNA polymerase, secondly, deamination of bases due to hydrolysis, and thirdly, thymidine dimers resulting from UV irradiation. Most of these lesions are repaired efficiently to restore the original DNA sequence, hence, avoiding formation of a mutation. Two features at the level of DNA ensure that repair is relatively simple and performed with high fidelity. Describe these two features.

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Answers:

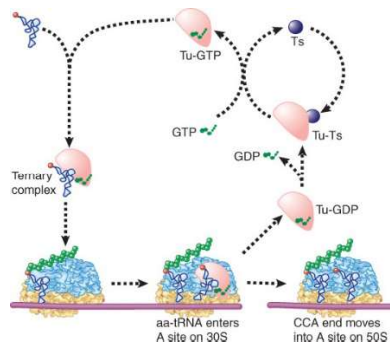
1) Lesion detection: The vast majority of lesions create a local spot on DNA where basepairing is abnormal. The repair machinery can survey DNA for the abnormal structures.

2) Repair: all lesions occur on a single strand; hence the other intact strand can be used as a template for flawless repair.

Question B.3. (6 %). Translation

Describe how Aminoacyl-tRNAs are loaded into the ribosome during translation elongation. Include a description of how the relevant transcription factors are recycled.

Answer: The aminoacyl-tRNA is bound to EF-Tu in the presence of GTP thus forming the ternary complex that enters the A site on the translating ribosome. EF-Tu is expelled from the ribosome under GTP hydrolysis. GDP is substituted by EF-Ts binding. The active form of EF-Tu is established by exchanging EF-Ts with GTP



Question B.4. (6 %). Eukaryotic activators.

Describe the mechanism by which a eukaryotic activator without an activating domain controls the expression of a gene.

Answer: The activator binds to a specific DNA sequence near the promoter. As the activating domain is missing, the activator recruits the basal apparatus to the promoter by a coactivator. Basal transcription factors bind to the promoter along with RNA polymerase, and initiates transcription.

Question B.5. (6 %). Bacteriophage development

Describe the lytic life cycle of a cos bacteriophage.

Answer: Attachment (adsorption) of the virus to a susceptible host cell

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Injection of virus DNA

Circularization by the host DNA ligase

Synthesis of circular virus DNA by theta replication.

Protein synthesis by cell metabolism as redirected by virus

Rolling circle replication and subsequent digestion into genome equivalents by the “cos” restriction enzyme

Assembly of capsids, and packaging of viral genomes into new virions (maturation)

Release of mature virions from host cell

Question B.6. (5 %). Evolution.

How do gene duplications promote evolution?

Answer: Most mutations are deleterious and would be selected against. If a gene is first duplicated, though, the duplicated copy is free to diverge, perhaps resulting in a desirable change, while the first copy retains the original function.

Part C. Large problem (25 %).

The answers should be written directly into the file in the allocated space, in one to three sentences.

In connection with the analysis of a bacterium isolated from fouling meat, the degradation of the amino acid serine was studied in more detail. Serine is converted by the enzyme Serine deaminase directly into pyruvate which is shunted into the TCA cycle. The gene encoding Serine deaminase (*sdeA*) was cloned and sequenced. A sequence analysis showed that the *sdeA* gene together with another gene *sdeB* constitutes an operon (Figure 1). It was also shown that *sdeB* encodes a membrane protein.

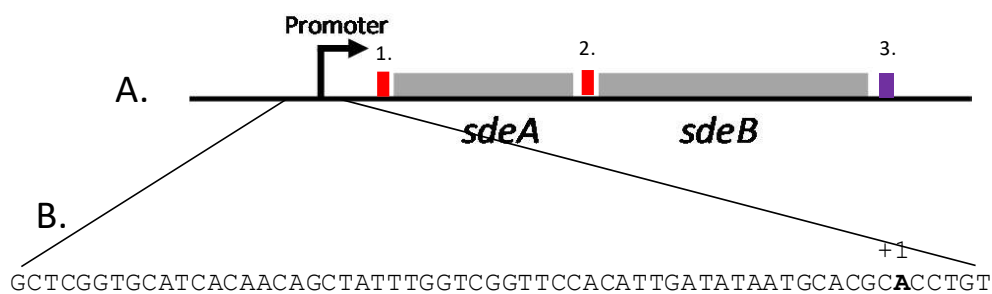


Figure 1. A. The genetic map of the *sde* operon. B. The *sdeAB* promoter sequence. The start of transcription is indicated as +1, and an inverted repeat identified in the sequence is underscored.

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Question 1 (2 %). In Figure 1A, three different genetic elements are shown (1, 2, and 3). What could they be?

Answer: 1 and 2 are the Shine-Delgarno sequences of the *sdeA* and *sdeB* genes, respectively. 3 could be the transcriptional terminator.

Question 2 (2 %). Indicate the position of the – 10 sequence in the promoter in Figure 1B on the sequence below:

GCTCGGTGCATCACAACAGCTATTTGGTCGGTTCCACATTGATATAATGCACGCACCTGT

Question 3 (3 %). What is the consensus of the -35 promoter sequence and what is the normal spacing between the -10 and -35? Can it be identified in Figure 1B?

Answer: TTGACA separated by 17 nucleotides from the – sequence. No, not present...

In a mutation analysis, three different mutants (BCA1, BCA2 and BCA3) that were no longer able to grow with serine as a source of carbon and energy were isolated following UV mutagenesis. Following sequencing analysis of the mutants, in BCA1 a missense mutation was found in *sdeA*, whereas in BCA2 a nonsense mutation was found in *sdeB*. In the third mutant, the entire *sdeAB* operon including its promoter was intact.

In a further study of the mutants, the content of *sdeA* and *sdeB* mRNA in the wild-type strain and in the three mutants was determined after growth in defined medium with and without added serine. In the table below, the relative mRNA amounts measured in the four strains are given.

Strain	Wildtype		BCA1		BCA2		BCA3	
		+ serin		+ serin		+ serin		+ serin
Relative <i>sdeA</i> mRNA amount	10	500	10	500	10	10	10	10
Relative <i>sdeB</i> mRNA amount	10	500	10	500	10	10	10	10

Question 4 (2 %). Which experiment that can be used to quantify a specific mRNA.

Answer: Reverse Transcriptase PCR or Northern blot

Question 5 (4 %). Explain why BCA1 is unable to utilize serine, and what kind of role the *sdeA* encoded protein plays in serine metabolism.

Answer: As SdeA may be the serine deaminase, and the missense mutation in BCA1 may result in an unfunctional enzyme.

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Question 6 (4 %). Explain why BCA2 is unable to utilize serine, and what kind of role the *sdeB* encoded protein plays in serine metabolism. Also explain why serine cannot induce transcription of the *sdeAB* operon.

Answer: As SdeB is a membrane protein, it may be responsible for the uptake the serine deaminase, and the nonsense mutation in BCA2 results in a unfunctional gene product. As serine no longer can enter the cell, it cannot induce transcription of the *sdeAB* operon.

Question 7 (4 %). Describe the effect of the mutation in BCA3? Give a possible explanation of the observed effect.

Answer: The expression of the operon is constitutive and low. It could be explained by a mutation in an activator located at another position of the chromosome.

Question 8 (4 %). In figure 1B, an inverted repeat is identified. Which role could that sequence play in the regulation of the *sdeAB* operon.

Answer: Binding site for an activator

Course 27026 – Molecular biology – E2020

Name:

Study number:

Technical University of Denmark

27026 Molecular Biology

Written exam, December 10th 2020,

Duration: 4 hours

All aid is allowed except personal communication

The exam set is divided into three parts, A, B and C. Part A is 20 Multiple Choice question; part B is 8 short questions, while part C is one large problem with 11 associated questions. A, B and C counts 30 %, 34 % and 36 % of the final grade.

Weighting: Answers must be written in the boxes, and answers should not exceed the provided space.

Part A. Multiple Choice (30 % - 1.5 % for each question)

The Multiple Choice part is accessed in the multiple Choice Module of DTU Inside. Each question counts 1.5 % of the total grade.

Part B. Short questions (34 %).

Problem B1. (6 points). Replication errors may lead to undesirable mutations. Replication errors are to a large degree prevented by the function of two different repair systems. One system function during replication and is contained within the DNA polymerase itself; the other, the mis-match repair system, functions after replication.

Describe the repair principles of the two repair systems. For the mis-match repair system, focus on the *E. coli* system and include an explanation on how *E. coli* determines which one of the two bases in a mismatched base pair should be replaced to restore the correct sequence.

Answer

Answer: **A) DNA polymerase:** If the **wrong nucleotide is added to the 3'-end it does not base-pair properly to the template**. This already makes addition of the next nucleotide difficult as the end is not positioned properly in the active site. **Next the loose interaction at the 3'-end allows for mobility** and the end may flip **into another site of the polymerase, which contains exonucleolytic activity**, and which **removes the 3'-base**. The 3'-end flips back into the polymerase site and **replication is resumed** now that basepair at the 3'-end is perfect. **B) The mis-match repair system:** i) **damage detection (MutS)**; ii) **removal of the wrong base - a single stranded stretch of DNA is removed including the wrong base. This occurs in two steps**; first a strand is nicked by **endonuclease MutH**, second, the strand with the wrong base is degraded by **RecJ or exonuclease VII** (both 5' to 3') or **exonuclease I** (3' to 5') dependent on the location of the mismatch relative to the nick. Degradation needs a helicase (**UvrD**) to separate strands. Iii) **The gap is filled in by DNA polymerase** and iv) the nick is sealed by **ligase**. The proper strand for repair can be chosen if the DNA is newly replicated and therefore hemimethylated. MutS in

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collaboration with MutL holds on to the mismatch while at the same time scanning strand for nearby hemi-methylated GATC sequences. **When a hemimethylated GATC site is identified, the mispaired base on the un-methylated “new” strand is removed.**

Explanations with the main features/principles in bold – which should be mentioned in the answer.

Problem B2. (4 points).

Describe briefly why telomeres are required to maintain linear chromosomes.

Answer

All DNA polymerases act in the 5' to 3' direction and needs a primer. The primer for synthesizing a chromosome is a small RNA synthesized by primase. The RNA primer is subsequently degraded and replaced with DNA by DNA polymerase at all internal locations. However, replication of the strand ending with a 5'-end cannot be completed by this method – per definition, no bases are available to serve as template for RNA primer synthesis for the very end of this strand. **As a consequence, chromosomes are shortened in each round of replication** (in fact, only one strand is shortened, and it takes a second round of replication to loose information on both strands). **To avoid (or delaying) losing coding/regulatory information at chromosome ends, they are capped with non-coding telomeres.** Telomeres are DNA repeats added to the 3'-end of chromosomes by telomerase. Telomerase brings in its own template, a small RNA molecule, to facilitate the extension of 3'-ends of chromosomes. The extended 3'-end is then made into double stranded DNA in the normal way by DNA polymerase (except the last telomere repeats, which remain single stranded for the same reasons described above).

Explanations with the main features/principles in bold – which should be mentioned in the answer.

Problem B3. (4 points).

Explain how cells ensure that the correct amino acid is attached to its corresponding tRNA

Answer

tRNA synthetase has two active sites. Amino acids larger than the right amino acid cannot be activated by adenylation because they do not fit in the synthetic site. Amino acids smaller than the correct amino acid are removed because they are able to enter the editing site, where it is hydrolyzed.

Problem B4. (4 points).

What factors are involved in promoter recognition of 1) the prokaryotic – 10 sequence, and 2) the eukaryotic TATA box. What is the state of the DNA during recognition in the two situations?

Answer

The sigma factor of the prokaryotic RNA polymerase binds to the single stranded DNA in the – 10 region, whereas the eukaryotic TBP in the TFIID complex interacts with the double stranded DNA in the TATA box.

Problem B5. (4 points).

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Explain briefly the principle of “epigenetic heritage” and give two examples hereof

Answer

DNA methylation pattern – preference of methylases towards hemimethylated DNA

Heterochromatin area – Protein self-assembly in the heterochromatin area

Prion (diseased tertiary structure of a protein) – The faulty protein fold, induces correctly folded proteins to re-fold into the diseased form

Problem B6. (4 points).

Describe briefly the principles of the major eukaryotic mRNA degradation pathways.

Answer: Two pathways exist. Both are initiated by de-adenylation of the poly A tail. Then either the exosome is recruited and degrades the mRNA from the 3' by a 3'-5' exonuclease or Dcp removes the 5' cap followed by 5'-3' exonuclease digestion.

Problem B7. (4 points).

Describe how rolling circle replication can generate multimers of replicated DNA.

Answer: Rolling circle replication is replication of one strand of a circular duplex template DNA. One of the two strands is nicked and the free 3' end is used as a primer to begin replication; the 5' end of the nicked strand is displaced. The template strand is circular, and thus replication can continue indefinitely, creating a multimeric single-stranded replication product.

Problem B8. (4 points).

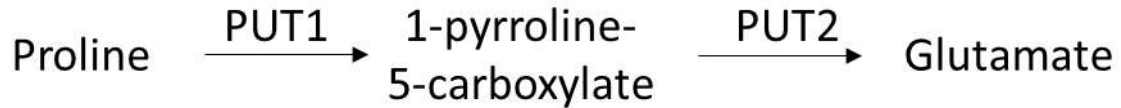
Explain how a small population can lead to a situation where a slightly deleterious allele that normally would be eliminated by weak negative selection, instead increase in frequency.

Answer

If the population is small in size, genetic drift (the random fluctuation of allele frequencies) may have a greater effect than selection. By chance, a variant that is weakly selected against may actually become more common.

Part C. Proline degradation in *Saccharomyces cerevisiae*. (36 %).

Proline can serve as a nitrogen source for the yeast *Saccharomyces cerevisiae* when preferred sources of nitrogen are absent from the growth medium. After uptake, proline is converted to glutamate that serves as nitrogen source. The conversion of proline to glutamate is dependent on proline oxidase and Δ -1-pyrroline-5-carboxylate dehydrogenase encoded by *PUT1* and *PUT2*, respectively.



PUT1 and *PUT2* are expressed only during conditions of poor nitrogen supply, and in a series of experiments, *PUT3* was shown to be involved in controlling expression of the two genes: first, the promoter from *PUT2* was fused to a *lacZ* gene, encoding β -galactosidase, and the construct was inserted into a number of yeast strains carrying different mutations. The strains were grown in a chemically defined medium supplied with either ammonia or proline. Compared to ammonia, proline is a poor nitrogen source. The specific β -galactosidase activity was measured in samples from the different experiments, and are shown in table 1.

Table 1. *PUT2* promoter activity in wildtype and mutants in proline degrading genes.

Strain name	Genotype	Specific β -galactosidase activity with medium containing	
		Ammonia	Proline
S. <i>cerevisiae</i>	<i>PUT2P:lacZ</i>	63	790
MB1	<i>PUT2P:lacZ; \Delta PUT1</i>	61	NG
MB2	<i>PUT2P:lacZ; \Delta PUT2</i>	67	NG
MB3	<i>PUT2P:lacZ; \Delta PUT3</i>	Below detection	NG
MB4	<i>PUT2P:lacZ; PUT3 (N914I)</i>	768	745

NG: No growth; Δ : Deletion. *PUT3* (N914I): The asparagine residue at position 914 substituted with isoleucine. Specific β -galactosidase activity: nmol/(min*mg prot)

Question 1 (4 %). Explain why strain MB1 and MB2 cannot grow on proline.

Answer: Because proline is the sole nitrogen source, and it cannot be degraded to glutamate since either *PUT1* or *PUT2* is deleted. Hence no nitrogen for biosynthesis

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Question 2 (4 %). There can be several explanations to why MB3 cannot grow with proline as the sole source of nitrogen. Give one.

Answer: *PUT3* is encoding an enzyme responsible for the utilization of glutamate as nitrogen source, it could be a proline transporter, or it could be an activator of *PUT1* and *PUT2* transcription.

To further analyze the nature of the mutants, the *PUT1* and *PUT2* wild type genes were cloned into a plasmid where expression of the two genes was controlled by the galactose inducible *GAL1* promoter. This plasmid, pMB1, was transformed into *MB1*, *MB2* and *MB3*, and the resulting strains were grown as described above in the presence or absence of galactose. In the absence of galactose, the β -galactosidase activities were similar to the results in Table 1. However, in the presence of galactose all strains could grow on proline as the sole nitrogen source. The β -galactosidase activities were as shown in Table 2.

Table 2. *PUT2* promoter activity in mutants in proline degrading genes supplied with a plasmid expressing *PUT1* and *PUT2* wildtype genes from a *GAL1* promoter. Strains were grown in the presence and absence of galactose.

Strain name	Genotype	Specific β -galactosidase activity with medium containing			
		Ammonia		Proline	
		- gal	+ gal	- gal	+ gal
MB1/pMB1	<i>PUT2P::lacZ; ΔPUT1, GALP1::PUT1::PUT2</i>	63	65	NG	757
MB2/pMB1	<i>PUT2P::lacZ; ΔPUT2, GALP1::PUT1::PUT2</i>	62	67	NG	761
MB3/pMB1	<i>PUT2P::lacZ; ΔPUT3, GALP1::PUT1::PUT2</i>	Below detection	Below detection	NG	Below detection

NG: No growth; Δ : Deletion. – gal: No galactose; + gal: galactose added. Specific β -galactosidase activity: nmol/(min*mg prot)

Question 3 (3 %). Explain why MB1/pMB1 and MB2/pMB1 can grow on proline as sole nitrogen source in the presence of galactose.

Answer: Galactose induces the expression of *PUT1* or *PUT2* thus supplying the mutants with the missing enzymes. It is a simple complementation.

Question 4 (3 %). Based on the finding that MB3/pMP1 can grow on proline as sole nitrogen source in the presence, what kind of function could *PUT3* have?

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Answer: *PUT3* is encoding an activator of *PUT1* and *PUT2* transcription. If it is absent, *PUT1* and *PUT2* is not expressed. In the presence of pMB1 *PUT1* and *PUT2* are transcribed from the plasmid by the galactose inducible *GALP1* promoter.

To validate the function of *PUT3*, the wildtype *PUT3* gene was cloned into a plasmid under control of the galactose inducible *GAL1* promoter. The resulting plasmid pMB3 was introduced into MB3 and MB4, and the β -galactosidase activities were measured in strains grown in the absence and presence of galactose. The assay results are summarized in table 3.

Table 3. *PUT2* promoter activity in mutants in proline degrading genes supplied with a plasmid expressing the *PUT3* wildtype gene from a *GAL1* promoter.

Strains were grown in the presence and absence of galactose.

Strain name	Genotype	Specific β -galactosidase activity with medium containing			
		Ammonia		Proline	
		- gal	+ gal	- gal	+ gal
MB3/pMB3	$\Delta PUT3$, <i>GALP1::PUT3</i>	Below detection	67	NG	728
MB4/pMB3	<i>PUT3</i> (N914I), <i>GALP1::PUT3</i>	726	737	743	765

NG: No growth; Δ : Deletion; *PUT3* (N914I): The asparagine residue at position 914 substituted with isoleucine. – gal: No galactose; + gal: galactose added.

Specific β -galactosidase activity: nmol/(min*mg prot)

Question 5 (3 %). Explain the phenotype of MB3/pMB3.

Answer: *PUT3* is encoding an activator. In MB3 it is deleted, and thus unable to degrade proline. If galactose is added, the *PUT3* gene is expressed from the *GALP1* promoter thus facilitating the expression of *PUT1* and *PUT2* reestablishing the ability to degrade proline.

Question 6 (3 %). Explain the phenotype of MB4 and MB4/pMB3.

Answer: The mutation (N914I) is resulting in an activator that is locked in its activating conformation. The mutation is dominant; the presence of a wildtype allele will not restore the wildtype phenotype.

The galactose regulatory system is a classical and well described system in yeast, and used here only to investigate the proline regulatory system.

Question 7 (2 %). The regulation of the *GALP1* promoter is dependent on three different regulatory proteins; which?

Answer: *GAL3*, *GAL4* and *GAL80*

To substantiate the previous findings, the expression of *PUT1* and *PUT2* was determined in a Northern blot experiment. RNA from *S. cerevisiae* wild-type and MB3 ($\Delta PUT3$) was extracted after

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growth in minimal media with ammonia or proline as sole nitrogen source. *PUT1*, *PUT2* and *ACT1* DNA was used as probes. *ACT1* is a constitutively expressed gene. The result is shown in figure 1.

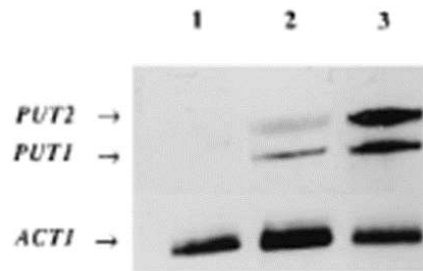


Figure 1. Northern blot with probes hybridizing to *PUT1*, *PUT2* and *ACT1* mRNA. Lane 1: MB3 grown on ammonia; Lane 2: Wildtype grown on ammonia; Lane 3: Wildtype grown on proline.

Question 8 (4 %). In figure 1, Why was an *ACT1* probe included, and what can be concluded from the *ACT1* band intensity?

Answer: To assess the amount of total RNA added to each lane. It shows that in lane 2 the total amount of RNA loaded was higher, meaning the two *PUT* bands are relatively lower.

Question 9 (3 %). Why are there no *PUT1*, *PUT2* bands in lane 1?

Answer: *PUT3* is indeed an activator, and since it is deleted, *PUT1* and *PUT2* are not expressed.

Question 10 (3 %). Why do the *PUT1* and *PUT2* bands have different intensities in lane 2 and 3. Is the difference in intensities consistent with the previous findings?

Answer: It signifies the amount of *PUT1* and *PUT2* is much higher when cells are growing on proline as nitrogen source. Is it in accordance with the observations in table 1, 2 and 3?

Figure 2 shows the genetic map of the *PUT1* promoter region.

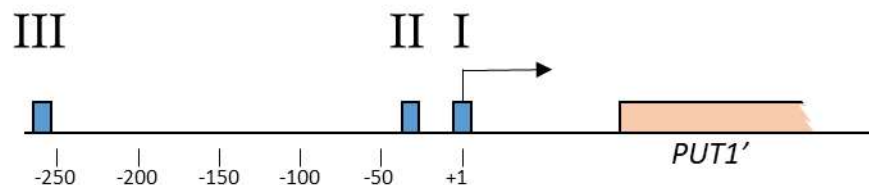


Figure 2. The genetic map of the *PUT1* promoter region inclusive the first part of the *PUT1* open reading frame. The numbers are

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relative to the transcriptional start site signified by the arrow. The blue boxes indicate the three genetic elements I, II and III.

Question 11 (4 %). What are the three blue boxes I, II and III in Figure 2 representing?
Answer: I: Inr, II: TATA box, III: UAS for PUT3 binding (Enhancer is also OK).

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Exam number:

27026 Molecular Biology

Written exam, December 8th 2021,

Duration: 4 hours

All aid is allowed except personal communication and internet access.

Answers must be written in the boxes in the word document, and not exceed the provided space. Remember to write exam number at the top. Upload your answers as a pdf file.

The exam is divided into three parts, A, B and C. Part A is 20 Multiple Choice questions; part B is 8 short questions, while part C is one large problem with 10 associated questions. A, B and C counts for 30 %, 32 % and 38 % of the final grade.

The Multiple-Choice part is accessed separately in the Multiple-Choice Module at DE. Each question counts 1.5 % of the total grade.

Part A. Multiple-Choice (30 % - 1.5 % for each question)

The Multiple-Choice part is accessed in the Multiple-Choice Module at DE.

Part B. Short questions (32 %).

Problem B1. (4 points).

Why does a high-fidelity DNA polymerase contain a 3'-OH to 5'-OH exonuclease activity? How do cells benefit from this activity?

Answer

Problem B2. (4 points).

Why is it necessary to decatenate e.g. the loops of eukaryotic chromosomes, and what enzyme is responsible for decatenation of eukaryotic chromosomes?

Answer

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Problem B3. (4 points). Translation initiation in eukaryotes.

Describe the role of the three listed initiation factors in eukaryotic translation and state which complex they are part of.	
eIF4E	
eIF1	
eIF3	

Problem B4. (4 points). Eukaryotic RNA polymerases

RNA polymerase II carboxyterminal domain (CTD) is characterized by several YSPTSPS-repeats, that are phosphorylated and dephosphorylated during transcription. The amino acids in the repeats are numbered from 1 to 7. Answer the following four questions:	
1. Which amino acid(s) is/are phosphorylated? 2. What is the phosphorylation state when the RNA polymerase is recruited by TFIID to the promoter? 3. One of the phosphorylations is coupled to splicing and polyadenylation. Which? 4. How is phosphorylation/dephosphorylation coupled to transcription termination?	
Answer 1. 2. 3. 4.	

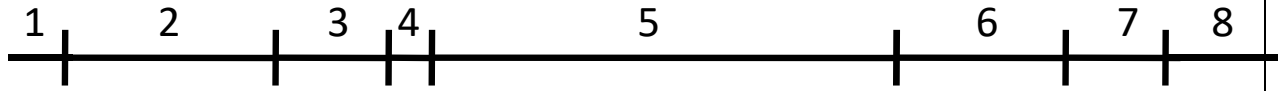
Problem B5. (4 points). Shuttle vectors.

What is a shuttle vector? Describe the genetic elements that must be present in a shuttle vector intended for use in both <i>Escherichia coli</i> and <i>Saccharomyces cerevisiae</i> .	
Answer	

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Problem B6. (4 points). mRNA processing

The drawing below shows the pre-mRNA from a eukaryotic gene. Element 1 encodes a signal sequence, whereas 2, 4, 6, and 8 are exons, and 3, 5, and 7 are introns. Describe the mature mRNA including the 3' and 5' end modifications. In which of the 8 elements will you find the start codon?



Answer:

Problem B7. (4 points).

Describe and name two mechanisms used by host cells to defend themselves against viral attacks.

Answer:

Problem B8. (4 points).

Describe one mechanism by which bacteria can acquire resistance against antibiotics?

Answer:

Part C. Regulation of gene expression in prokaryotes (38 %).

The expression of proteins A and B from *genA* and *genB* in *E.coli* is induced by adding an inducer to the growth medium. The enzymatic activity of A and B was assayed in the absence or presence of this inducer, and shown in table 1.

Table 1. The enzymatic activities of A and B assayed in the absence or presence of inducer.

Inducer addition	Enzymatic level protein A (units/mg)	Enzymatic level protein B (units/mg)
-	10	15
+	200	300

Question 1 (2 %). *genA* and *genB* were experimentally proven to constitute an operon. What is the criterion for *genA* and *genB* being in the same operon?

Answer:

Question 2 (2 %). How can you experimentally verify that two genes are part of the same operon?

Answer:

The genetic map of the *genAB* operon is shown in figure 1.

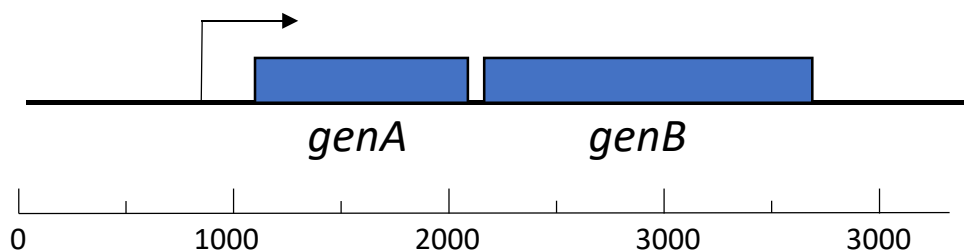


Figure 1. The genetic map of the *genAB* operon. An arrow indicates the approximate position of the promoter. A marker in basepairs below the map shows the size of the operon.

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To identify the promoter, a set of transcriptional fusions between *genA* and *lacZ* were constructed, and introduced into *E. coli*. The cloned fragment ranges from 0 to 1200 base pairs according to the scale in Figure 1.

Question 3 (2 %). Before introducing the transcriptional fusions into the *E. coli* strain, the researcher inactivated one locus in the *E. coli* genome. Which *E. coli* gene needs to be inactivated in order to use the *lacZ* reporter fusions to measure promoter activity?

Answer:

After establishment of the transcriptional fusion, the β -galactosidase activity was assayed after growth in medium in the absence and presence of the inducer. The results are shown in Table 2.

Table 2. Enzymatic assay of *E. coli* strain carrying a *genA::lacZ* fusion

Inducer addition	β -galactosidase activity (units/mg)
-	50
+	200

Question 4 (5 %). Comparing the induction of the enzymatic activities

Calculate i) the fold of induction in the enzymatic activities of A and B (Table 1), and ii) the fold of induction found in the transcriptional fusion (Table 2)

Answer:

Induction fold:

i) A/B enzyme assay =

ii) *genA::lacZ*: =

If you find a difference, how would you explain it?

Answer:

To further characterize the regulation of the *genAB* operon, the *E. coli* strain harbouring the *genA::lacZ* fusion was subjected to transposon mutagenesis, and strains with altered *lacZ* expression were isolated. Two transposon mutants – Mut1 and Mut2 - were obtained. In Mut1 the transposon was inserted into a gene (*genP*) encoding a protein with several putative α -helices of hydrophobic

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nature. In Mut2, the transposon had inserted into *genR* encoding a protein with a characteristic Helix-Turn-Helix motif. Next, the β -galactosidase activity in Mut1 and Mut2 was determined in the absence and presence of the inducer. The measured activities are listed in Table 3.

Table 3. Enzymatic assay of *E. coli* strain carrying a *genA::lacZ* fusion

Strain	Mutant type	Inducer addition	β -galactosidase activity (units/mg)
Wildtype		-	50
		+	200
Mut1 (<i>genP</i>)	Hydrophobic α -helixes	-	50
		+	50
Mut2 (<i>genR</i>)	Helix-Turn-Helix motif	-	200
		+	200

Question 5 (4 %).

What does the presence of hydrophobic α -helixes in a protein suggest about its function?

Answer:

How does the mutation in Mut1 alter transcription from the *genA*-promoter? Suggest a function for the protein mutated in Mut1. Briefly explain why.

Answer:

Question 6 (4 %).

What does the presence of a Helix-Turn-Helix motif in a protein suggest about its function?

Answer:

How does the mutation in Mut2 alter transcription from the *genA*-promoter? Suggest a function for the protein mutated in Mut2. Briefly explain why.

Answer:

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By PCR, *genR* was amplified from wildtype chromosomal DNA, and cloned into a plasmid to give pGenR⁺. This plasmid was subsequently transformed into the Mut2 strain. The resulting strain was assayed for β -galactosidase activity in the absence and presence of the inducer.

Question 7 (4 %). Predict the result of the β -galactosidase assay with the Mut1 carrying pGenR⁺

Answer:

Activity of Mut1//pGenR⁺ without inducer:

Activity of Mut1//pGenR⁺ with inducer:

To further characterize the regulation of *genAB*, the promoter was substituted with a constitutive promoter, leaving the leader unchanged. The enzyme activity of A was measured (Table 4).

Table 4. The enzymatic activities of A and B assayed in a strain where the *genAB* operon is expressed from a constitutive promoter. Strain were grown either in the absence or presence of the inducer.

Inducer addition	Enzymatic level protein A (units/mg)
-	40
+	200

Question 8 (5 %). Compare the induction fold of the enzymatic activities

Calculate i) the fold of induction in the enzymatic activity of A (Table 1) and ii) the fold of induction found in the presence of a constitutive promoter (Table 4)

Answer:

Induction fold:

i) A enzyme assay wildtype promoter =

ii) A enzyme assay constitutive promoter =

If you find a difference, how would you explain it?

Answer:

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The *genAB* leader can fold into two mutually exclusive structures shown in figure 2.

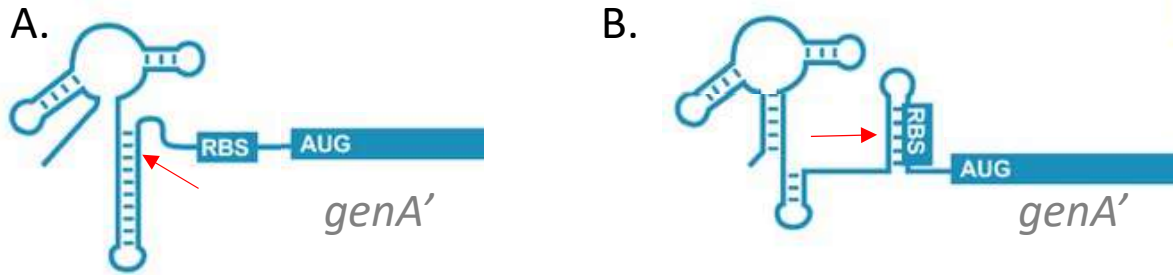


Figure 2. The untranslated leader in the *genAB* operon was predicted to be capable of folding into two alternative structures shown in A. and B. The Shine-Dalgarno sequence and the start codon are noted as RBS and AUG, respectively. The first part of the *genA* open reading frame is shown as a blue box. The red arrow points to the mutated nucleotide in Mut3. (See description of Mut3 in the text below).

Question 9 (5 %). Comparing the structures of the leader (Figure 2)	
Structure A will result in higher translation initiation frequencies. Why?	Answer:
Which of the two structures will be stabilized by binding of the inducer? Explain why.	Answer:

A third mutant, Mut3, was obtained after UV mutagenesis of the strain expressing *genAB* from a constitutive promoter. The mutation in Mut3 resulted in constitutively high expression of *genAB*. The mutation in Mut3 introduced a base shift in the *genAB* leader at the position marked with a red arrow in figure 2.

Question 10 (5 %). Explain why Mut3 has a higher expression of <i>genAB</i> in the absence of the inducer.
Answer:

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Exam number:

27026 Molecular Biology

Written exam, December 8th 2022,

Duration: 4 hours

All aid is allowed except personal communication and internet access.

Answers must be written in the boxes in the word document, and not exceed the provided space. Remember to write exam number at the top. Upload your answers as a pdf file.

The exam is divided into three parts, A, B and C. Part A is 20 Multiple Choice questions; part B is 8 short questions, while part C is one large problem with 10 associated questions. A, B and C counts for 30 %, 32 % and 38 % of the final grade.

The Multiple-Choice part is accessed separately in the Multiple-Choice Module at DE. Each question counts 1.5 % of the total grade.

Part A. Multiple-Choice (30 % - 1.5 % for each question)

The Multiple-Choice part is accessed in the Multiple-Choice Module at DE.

	correct answer
1. An expression vector... (choose the right answer)	
A) carries a reporter gene B) moves genes from yeast cells to bacteria C) produces an enzyme from a cloned gene D) inhibits restriction enzymes E) encodes at least two origins of replication	C
2. Which technology cannot be used for RNA quantification	
A) Southern blot B) Northern blot C) Reverse transcriptase PCR D) RNAseq E) Dot-blot	A
3. Genomes: An intron is a DNA element which	
A) in prokaryotes are used to separate promoter elements B) in prokaryotes are used to separate exons in the origin of replication C) in prokaryotes are used to separate exons in a gene D) in eukaryotes are used to separate exons in a gene E) in eukaryotes are used to separate telomeric repeats	D

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4. Replication: The template for synthesizing telomeres are... (choose the right answer)	
A) DNA embedded in telomerase B) RNA embedded in telomerase C) tRNA embedded in telomerase D) Exon DNA embedded in telomerase E) No template; telomerase does not require a template	B
5. Replication: During DNA replication primase synthesizes... (choose the right answer)	
A) A short stretch of RNA acting as a substrate for DNA polymerases B) Telomeric DNA C) Centromeric DNA D) Lagging strand DNA E) Leading strand DNA	A
6. Repair: One pathway repairing pyrimidine dimers is coupled to transcription. As a consequence... (choose the right answer)	
A) repair of pyrimidine dimers in transcribed DNA strands are repaired faster than those that are present in the rest of the genome B) repair of pyrimidine dimers in the non-transcribed DNA strands are repaired faster than those that are present in the rest of the genome C) repair of pyrimidine dimers in the telomers are repaired faster than those that are present in the rest of the genome D) repair of pyrimidine dimers in the centromeric are repaired faster than those that are present in the rest of the genome E) repair of pyrimidine dimers in mRNA are repaired faster than those that are present in the genome	A
7. Repair: A key enzyme in repair of a DNA double strand break by non-homologous end-joining is ... (choose the right answer)	
A) Holliday Junction resolvase B) The recombinase Rad51 C) Telomerase D) Primase E) DNA ligase	E
8. The following DNA sequence (coding strand) encode an <i>E. coli</i> Sigma70 promoter. 5' -ATGTCCAACCCATATAATTGACATAACGTAAGGTTGTTGTTATAATCATACGATCCAGTAGG-3'	
After identifying the promoter, what would be the sequence of the first six nucleotides in the RNA?	
A) 5'-AUCCAG-3' B) 5'-AUGUCC-3'	A

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C) 5'-AGTAGG-3' D) 5'-ACGUAA-3' E) 5'-CCUACU-3'	
9. Which statement addressing the eukaryotic RNA polymerase II and its promoter is true?	
A) RNA polymerase II synthesises the primers during DNA replication B) Only RNA polymerase II with a sigma factor can initiate transcription C) RNA polymerase II is active primarily in the cytoplasm D) Transcription Factor TF _{II} D recognizes Enhancers E) TATA less promoters are also dependent on the TATA Box Binding protein	E
10. Which statement regarding the exon junction complex (EJC) is false ?	
A) EJC assembles at exon-intron junctions B) EJC assists in mRNA transport from the nucleus to the cytoplasm C) EJC includes a ribonuclease D) EJC is found in both the cytoplasm and the nucleus E) All proteins are part of the EJC	C
11. Below you see the start of an open reading frame (in mRNA) in a wild-type and five mutants Met-Phe-Tyr-Gly Wildtype: ATG-UUU-UAC-GGC Mut1 : ATG-UUG-UAC-GGC Mut2 : ATG-UUU-UAA-GGC Mut3 : ATG-UUU-UAC-GGA Mut4 : ATG-UUA-UAC-GGC Mut5 : ATG-UUU-UCC-GGG	
Which statement is wrong ?	
A) Mut3 is silent B) Mut2 is a nonsense mutation C) Mut1 is a missense mutation D) Mut4 is silent E) Mut5 is a missense mutation	D
12. What is the role of eIF2 in translation	
A) Brings the initiator met-tRNA to the preinitiation complex B) Prevent the 40S and 60S subunits from joining during the early stages of initiation C) Binds to the 5'-CAP D) Acts as a RNA helicase E) Binds to the PolyA tail	A
13. In prokaryotic gene regulation, a mutation in a repressor binding site... (choose the right answer)	
A) leads to constitutive low gene expression B) leads to constitutive high gene expression C) can be complemented by a wildtype allele	B

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D) results in impaired binding of an inducer E) is lethal	
14. Prokaryotic gene regulation. Which of the following regulatory mechanisms does not control transcription?	
A) Repressor control B) Activator control C) Attenuation D) Shine-Dalgarno sequestering E) Anti-termination	D
15. Prokaryotic gene regulation. The gene <i>purF</i> is regulated by a transcriptional activator that uses metabolite M as a co-activator. Which statement is wrong ?	
A) In the presence of M, expression of <i>purF</i> is high B) If the activator is unable to bind to the activator binding site, <i>purF</i> expression is low C) Transcription of <i>purF</i> is induced if the activator is unable to bind M D) In the absence of M, expression of <i>purF</i> is low E) Transcription of <i>purF</i> will be constitutively high if a mutated activator binds in the absence of M.	C
16. Which of the following statements about true activators is false	
A) They need to access the nucleus to function B) They make direct contact with the basal transcriptional apparatus C) They can be sequestered in the cytoplasm by a repressor D) They affect the stability of the nascent mRNA E) They are involved in positive and negative controlled genes	D
17. Eukaryotic transcription regulatory proteins... (choose the right answer)	
A) Always dimerize upon interaction with DNA B) Are composed by a DNA-binding domain and a transcription activation domain C) Need a co-activator protein to make contact between DNA and the basal apparatus D) Are involved only in the positive regulation of gene expression E) Need a ligand to become active	B
18. Which of the following statements about miRNAs is false ?	
A) miRNAs are non-coding regulatory RNAs B) miRNAs need a carrier protein to find their target sequence C) miRNAs are processed by multiple nucleases D) miRNAs are partly located in the leader sequence E) miRNAs have multiple targets	D
19. A riboswitch is a regulatory segment located on the non-coding region of an mRNA that can adopt mutually exclusive structures by sensing...choose the right answer	
A) a specific tRNA abundance B) a protein C) a metabolite D) a mutation E) ssDNA <22 nucleotides	C
20. During the HIV virus infection cycle... (choose the right answer)	

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A) the viral RNA is transcribed by an RNA dependent RNA polymerase B) a host protease cleaves the Gag and Pol poly-proteins C) a host tRNA serves as a primer for the reverse transcriptase D) no mRNA splicing occurs, as there are no introns E) lysozyme is produced so that ultimately, the host cell is lysed	C
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Part B. Short questions (32 %).

Problem B1. (4 points).

The dam methylase plays an important role in replication of the *E. coli* chromosome. Briefly explain how *E. coli* use the dam methylase to keep the replication induced mutation rate low.

Answer: Methylation of the old DNA strand plays an important role in replication fidelity. If DNA polymerase inserts a wrong base and if the proof reading activity of the polymerase fails to correct it; then the cell still has a chance to correct the error as the wrong base on the new strand cannot pair correctly to the base on the old (template)-strand. The mismatch can be detected, by the mismatch repair system, which will remove the wrong base and insert the correct one. The mismatch repair system can choose the correct strand to repair, the newly synthesized strand, as long as it has not been methylated by the dam methylase. Hence, there is a window of opportunity controlled by the dam methylase where the cell can accurately fix replication errors.

Problem B2. (4 points).

Repair of DNA double strand breaks by homologous recombination can occur by two different pathways: the “classical DNA double strand break repair (DSBR) and by synthesis dependent strand annealing (SDSA). DSBR may lead to crossovers whereas SDSA does not. Briefly explain why only DSBR produces crossover products.

Answer

The DSBR pathway, but not the SDSA pathway, forms Holliday junctions. In DSBR, both ends interact with the homologous target repair sequence. This sets the stage for formation of double Holliday junctions. Resolution of the Holliday junctions may produce crossover (50 % chance). In SDSA no Holliday junctions are formed as only one end interacts with the target repair sequence. No intermediate in the SDSA repair process is able to produce crossovers.

Problem B3. (4 points).

Mobile genetic elements: Briefly explain why a DNA transposon (Class II) is flanked by a short, direct repeat of host DNA

Answer: When a DNA transposon (Class II) integrates into the host genome, its transposase asymmetrically cuts a site on a host chromosome to produce a DNA double stranded break with staggered ends. The Transposon then inserts the transposon into the DNA DSB via ligation to the overhanging ends of the DNA DSB in a reaction that produces two small single stranded gaps of complementary host DNA. The gaps will be filled in by DNA polymerase and sealed by ligase. This reaction produces a short direct repeat of host DNA flanking the DNA transposon.

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Problem B4. (4 points).

The following elements are found in eukaryotic and prokaryotes: Allocate them according to where they are found (some may belong to both groups).	
Enhancer, Shine-Dalgarne sequence, TATAbox, - 35 sequence, 5'-CAP, GU-AG splice site, Initiator, Operator, - 10 Sequence, Pribnow box, Intron, open reading frame, Leader sequence, AUS, Silencer, TF _{II} D	Prokaryotes: Shine-Dalgarne sequence, - 35 sequence, Operator, - 10 Sequence, Pribnow box, Intron, Open reading frame, Leader sequence Eukaryotes: Enhancer, TATAbox, 5'-CAP, GU-AG splice site, Initiator, Intron, Open reading frame, Leader sequence, AUS, Silencer, TF_{II}D

Problem B5. (4 points).

Describe briefly the following steps in the spliceosome mediated mRNA splicing:	
Initial recognition of the 5' splice site	The U1 snRNP binds to the 5' splice site through RNA-RNA base pairing
Bringing the 5' and 3' splice sites close to each other	The U2AF, the 63/65 protein and Branch-Point-Protein bound to the 3' splice site and U1 at the 5' splice site are brought together by the SR protein
Formation of the catalytic center	U6 basepairs with 5' splice site and U2, which is also pairing with the branch site. U5 interacts with both exons

Problem B6. (4 points). Chromatin remodeling

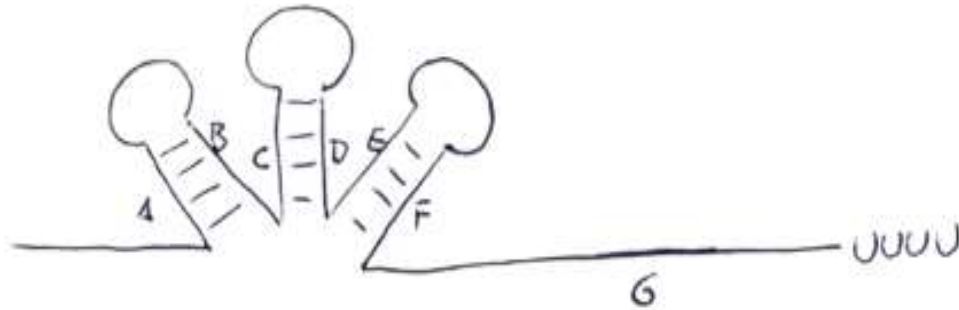
Describe how chromatin remodeling influences gene transcription. List three different modifications of histones and describe how the modifications changes gene transcription.
Chromatin remodelling is an active process which consists on the posttranslational modification of histones, modifying the packing of DNA around them and making them more or less available for transcription. - Histone acetylation is associated with transcriptional activation - Histone methylation can be associated with both transcriptional activation and repression

- Histones phosphorylation is associated to transcription, repair, chromosome condensation, and cell-cycle progression.

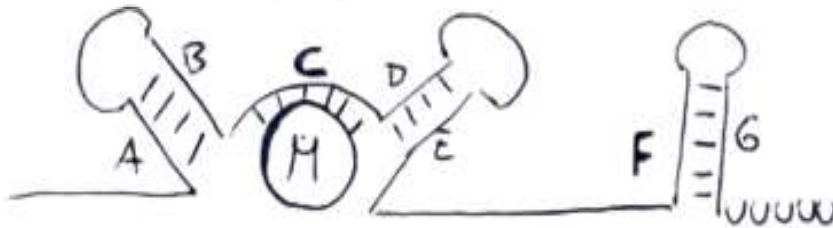
Problem B7. (4 points). Riboswitches

The figure shows the two alternative folds of a riboswitch in the absence and presence of the ligand M. The letters A to G represents the different parts of the RNA involved in structure formation and ligand binding.

A. Without ligand (-M)



B. With ligand (+M)



Different fragments of the riboswitch were fused to a *lacZ* reporter gene, and the β -galactosidase activity determined in the absence and presence of the ligand.

Describe briefly the mode of action of the riboswitch with respect to gene expression

Ligand absent: In the absence of the ligand, the riboswitch adopts an antiterminator structure, leading to the expression of the controlled genes.

Ligand present: In the presence of the ligand, the riboswitch binds it, changing their structure and forming a terminator structure that impairs the expression of the controlled genes.

Fill the following table with the expected β -galactosidase activities in the presence and

Sequence inserted	-M	+M
Wild type (A to G)	100	1
ΔF (deletion of F fragment)	100	100

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absence of M, assuming that the repressed activity is 1, and the induced is 100.	$\Delta A-E$ (deletion of fragments A to E)	1	1	

Problem B8. (4 points).

Briefly describe how the <i>Escherichia coli</i> bacteriophage T7 regulates gene expression to ensure correct timing in phage development?
The first genes are expressed by the host RNA polymerase. Among these genes are the T7 RNA polymerase and a protein that silences the host RNA polymerase. When these two proteins are synthesized in adequate amounts, the middle and late genes will be expressed by the T7 RNA polymerase

Part C. The utilization of maltose in *Saccharomyces cerevisiae* (38 %).

Maltose is a disaccharide consisting of two glucose molecules. *Saccharomyces cerevisiae* can take up maltose from the environment through a maltose transporter encoded by *MALT*. After uptake, maltose is hydrolysed into two glucose molecules by the *MALS* encoded maltase enzyme. Together with a regulatory gene *MALR*, the genes are organized in a gene cluster as shown in figure 1.

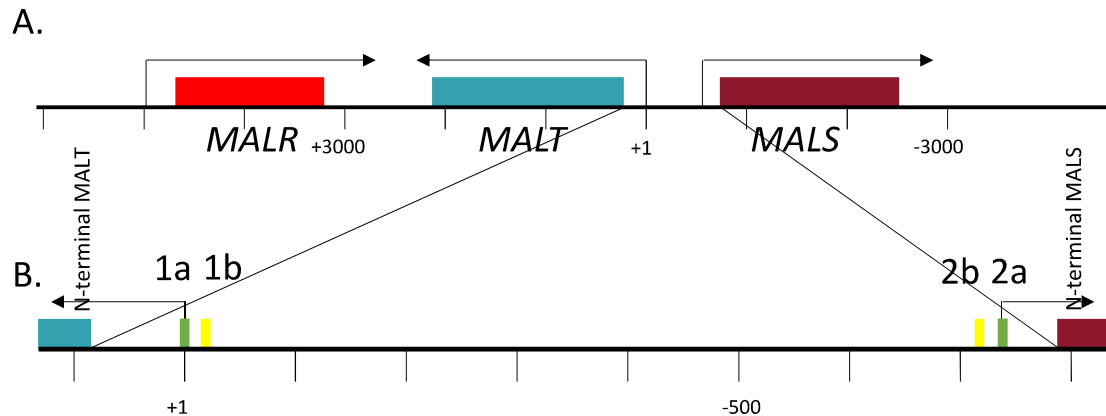


Figure 1. The *MAL* gene cluster of *Saccharomyces cerevisiae*. The numbers are relative to the transcriptional start site of *MALT*. The arrows show the transcriptional start sites. A. Genetic organization of the *MAL* gene cluster. B. Detailed map of the *MALT* and *MALS* promoter region. The green and yellow coloured boxes (1a, 1b, 2a and 2b) indicate the genetic elements required for transcription initiation.

Question 1 (2 %). The *MAL* promoters.

What are the boxes 1a, 1b, 2a and 2b in figure 1 representing?
1a: Inr (initiator)
1b: TATA boxes
2a: Inr (initiator)
2b: TATA boxes

To analyse regulation of *MAL* gene expression, *S. cerevisiae* wildtype and *MALR* deletion strains were grown in media supplied with either maltose, glucose or maltose as carbon sources. The *MALR* deletion strain was unable to grow with maltose as the sole carbon source, and for this reason a combination of glycerol and maltose was used in some experiments. mRNA was extracted and used in a Northern experiment, with either a *MALT* or *MALS* specific probes. The result is shown in figure 2.

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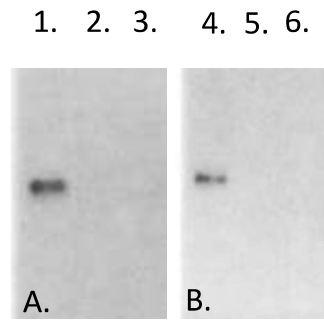


Figure 2. Northern blot analysis of the *S. cerevisiae* wildtype and *MALR* deletion strains grown in different media. Probes were either *MALT* (A) or *MALS* (B). Lane 1 and 4: Wildtype grown in Maltose; Lane 2 and 5: Wildtype grown in Glucose; Lane 3 and 6: Δ *MALR* grown in Maltose plus Glycerol.

Question 2 (4 %). What can you conclude with respect to *MALT* expression by comparing lane 1 and lane 2 in figure 2?

Answer:

Maltose induces expression of *MALT*

Question 3 (4 %). What can you conclude with respect to the effect of *MALR* on *MALT* expression by comparing lane 1 and lane 3 in figure 2?

Answer:

MALR is required for *MALT* expression

Question 4 (2 %). What kind of protein could *MALR* encode?

Answer:

MALR could encode an activator

Question 5 (2 %). What can you conclude with respect to the regulation of *MALT* and *MALS* expression by comparing Blot A and B?

Answer:

MALT and *MALS* are regulated by the same mechanism

Question 6 (4 %). Why was the *MALR* deletion strain unable to grow on Maltose?

Answer:

Since the *MALS/MALT* genes need *MALR* for expression, there are neither a Maltose transporter nor a Maltase enzyme produced. Hence, maltose cannot be utilized.

To further analyse the regulation of the *MALS* gene expression, different fragments were inserted in front of a promoter-less β -galactosidase gene and inserted into *S. cerevisiae* wildtype. The strains were grown in media supplied with either glucose or maltose, and the β -galactosidase activities assayed. The different constructs and the corresponding β -galactosidase activities are shown in figure 3.

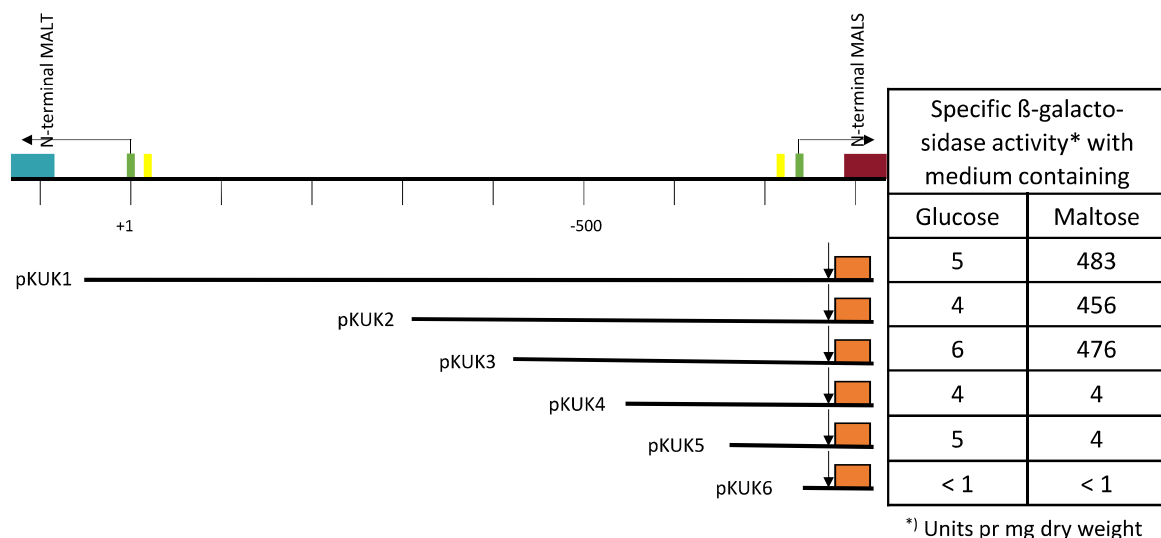


Figure 3. Deletion analysis of the *MAL* gene cluster of *Saccharomyces cerevisiae*. The numbers are relative to the transcriptional start site of *MALT*. The arrows signify the transcriptional start sites. The green and yellow coloured boxes indicate the genetic elements required for transcription initiation. Below are the different fragments used for construction of reporter plasmids indicated. The orange box represents the start of the *lacZ* gene used as reporter. The fusion point is indicated by an arrow. The table shows the specific β -galactosidase activities in the different strains harbouring the indicated plasmids.

Question 7 (2 %). Why is the activity from pKUK6 below detection?

Answer:

The TATA box is missing

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Question 8 (2 %). Are the results with pKUK1 in accordance with the findings from the Northern blot analysis? Briefly state why.

Answer:

Yes – maltose induces expression in both the Northern and in the *lacZ* fusion.

Question 9 (4 %). What could account for the finding that pKUK4 and pKUK5 have constitutive low β -galactosidase activity and pKUK3 shows high activity in the presence of maltose compared to glucose?

Answer:

An UAS sequence is missing in pKUK4 and 5

Question 10 (4 %). If the *MAL* fragment in pKUK1 was inserted in the opposite direction so that the *lacZ* gene was fused to the *MALT* promoter, how would you expect the β -galactosidase to be in presence of maltose compared to glucose?

Answer:

Approximately the same as in the original orientation.

The plasmids were also introduced in the Δ *MALR* mutant and grown in a media with either glucose or glycerol plus maltose as carbon source.

Question 11 (4 %). How would you expect the β -galactosidase activities to be – repressed, induced or undetectable?

Answer:

They will all have the repressed activity with both glucose and glycerol plus maltose –expect pKUK6 that still would have undetectable activities.

Finally, the wildtype strain harbouring pKUK1 was also analysed after growth a medium with both maltose and glucose. The β -galactosidase was found to be low. How would you explain this observation?

Question 12 (4 %). Finally, the wildtype strain harbouring pKUK1 was also analysed after growth a medium with both maltose and glucose. The β -galactosidase was found to be low. How would you explain this observation?

Answer:

The low expression is due to carbon catabolite expression

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Study number:

27026 Molecular Biology

Written exam, December 7th 2023,

Duration: 4 hours

All aid is allowed except personal communication and internet access.

Answers must be written in the boxes in the word document, and not exceed the provided space. Remember to write exam number at the top. Upload you answers as a pdf file.

The exam is divided into three parts, A, B and C. Part A is 20 Multiple Choice questions; part B is 8 short questions, while part C is one large problem with 10 associated questions. A, B and C counts for x %, y % and z % of the final grade.

The Multiple-Choice part is accessed separately in the Multiple-Choice Module at DE. Each question counts w % of the total grade.

Part A. Multiple-Choice (30 % - 1.5 % for each question)

The Multiple-Choice part is accessed in the Multiple-Choice Module at DE.

	correct answer
1. In a PCR experiment amplifying a specific DNA sequence, which parameter significantly influences the stringency of primer annealing?	
A) Volume of the PCR reaction B) The concentration of deoxribonucleotides (dNTPs) C) Annealing temperature D) DNA polymerase type E) Template DNA concentration	C
2. What can be used as a probe in a Southern blot?	
A) Sheared chromosomal DNA B) A fluorescence-labelled antibody C) A radioactive labeled oligo deoxyribonucleotide D) tRNA E) mRNA	C
3. Which process benefits from the presence of telomeres?	
A) Replication of centromeres B) Replication of chromosome ends C) Replication of plasmids D) Replication of exons E) Replication of introns	B
4. Replication of a chromosome requires RNA primers. In eukaryotes , what enzyme activity removes these primers to avoid RNA in the chromosomes?	
A) RNA exonuclease B) DNA Polymerase I exonuclease activity	E

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C) DNA Polymerase II exonuclease activity D) DNA Polymerase III exonuclease activity E) Flap endonuclease activity	
5. Why do you need topoisomerase activity in front of the replication fork?	
A) To remove replication-induced positive DNA supercoils B) To remove replication-induced negative DNA supercoils C) To remove the RNA primer D) To fuse Okazaki fragments E) To remove telomeres	A
6. A yeast strain with defective telomerase may form circular chromosomes. Which DNA repair pathway is required to form the circles?	
A) Base-excision repair B) Nucleotide-excision repair C) Mismatch-excision repair D) Non-homologous End-joining E) Homologous recombination	D
7. Heteroduplex may lead to a specific event on DNA, which one?	
A) Gene conversion B) DNA crossing over C) DNA double-strand break formation D) DNA single-strand break formation E) Loss of telomeres	A
8. Some plasmids may transfer from one <i>E. coli</i> cell to another by conjugation: But what is actually transferred from cell to cell?	
A) a linear single-stranded DNA molecule B) a linear double-stranded DNA molecule C) a circular single-stranded DNA molecule D) a circular double-stranded DNA molecule E) a linear RNA molecule	A
9. What is needed before a eukaryotic gene obtained by Reverse transcriptase PCR can be expressed in a bacterium?	
A) Deletion of introns B) Removal of the 5'-Cap C) Removal of the polyA- tail D) Changing the U residues to T's E) Addition of Shine-Dalgarno sequence	E
10. Which of the factors/events is NOT involved in eukaryotic transcription	
A) Phosphorylation of the Carboxy-terminal Domain (CTD) of the RNA polymerase B) TATA-box binding protein C) The initiator Inr D) The 43S preinitiation complex E) Transcription factor TF _{II} D	D
11. In eukaryotic cells, how is transcription termination typically achieved in mRNA synthesis?	
A) Polyadenylation signal recognition	A

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B) Release factor interaction C) Hairpin loop formation D) Rho protein termination E) tRNA dissociation	
12. Which combination of snRNAs within the spliceosome forms the catalytic center responsible for the transesterification reactions leading to intron excision during pre-mRNA splicing?	
A) U1 and U4 snRNAs B) U2 and U5 snRNAs C) U2 and U6 snRNAs D) U4 and U6 snRNAs E) U5 and U6 snRNAs	C
13. How is GTP hydrolysis associated with translation elongation?	
A) GTP hydrolysis is not required for translation elongation only initiation B) GTP hydrolysis is required for binding of the amino acid to the tRNA C) GTP hydrolysis is required for the formation of the peptide bond D) GTP hydrolysis is required for EF-Tu to dissociate from the ribosome E) GTP hydrolysis is required for binding of the release factor	D
14. Which cellular structure is primarily responsible for the assembly of amino acids into a polypeptide chain during translation?	
A) Ribosome B) Mitochondria C) Endoplasmic reticulum D) Golgi apparatus E) Lysosome	A
15. For the majority of DNA-binding proteins, the specificity of DNA binding is determined by direct contact with...	
A) Negatively charged phosphate backbone of DNA B) The major groove of DNA C) The minor groove of DNA D) The major and the minor groove of DNA at the same time E) No Direct contact with DNA, as binding requires specialized adaptors that interact with DNA	B
16. In a eukaryote cell, a repressor...	
A) always binds to the promoter preventing the binding of RNA polymerase II B) interacts with true activators only in the nucleus C) can act by sequestering true activators in the cytoplasm D) cannot interact with DNA E) respond only to environmental stimuli	C
17. In a prokaryote cell, a trans-regulatory element is...	
A) a non-coding DNA sequence B) always a protein that can be supplemented externally C) always encoded by the same operon which expression it regulate D) a diffusible molecule that can be supplemented externally	D

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E) a TATA-less promoter	
18. When the leader RNA of an operon can adopt terminator/antiterminator mutually exclusive structures...	
A) the terminator structure provokes the detachment of RNA polymerase II, interrupting the synthesis of the complete mRNA B) the operon is controlled by repression C) the antiterminator structure is always stabilized by the binding of a metabolite D) the stability of the mRNA molecules depends on the recruitment of RNAses by the RISC complex E) the antiterminator structure provokes the detachment of the RNA polymerase II, interrupting the synthesis of the complete mRNA.	A
19. Which of the following statements about histone remodelling complexes is TRUE	
A) are highly sequence-specific B) its activity always condenses the chromatic around the histones C) histone acetylation is associated with repression of gene expression D) always contain an ATPase catalytic subunit E) histone methylation is always associated with repression of gene expression	D
20. Regarding the use of viruses as vectors for gene delivery to eukaryotic cells, which of the following statements is FALSE	
A) Lentiviruses are suitable for stable gene expression B) Adenoviruses are suitable for transient gene expression C) Adenoviruses switch between lytic and lysogenic forms D) Lentiviruses integrate their genetic material within the genome host E) Adenoviruses have a broad cell tropism	C

Part B. Short questions (34 %).

Problem B1. (4 %). Genomes

The end of the telomere is arranged as a T-loop as shown in the drawing below:



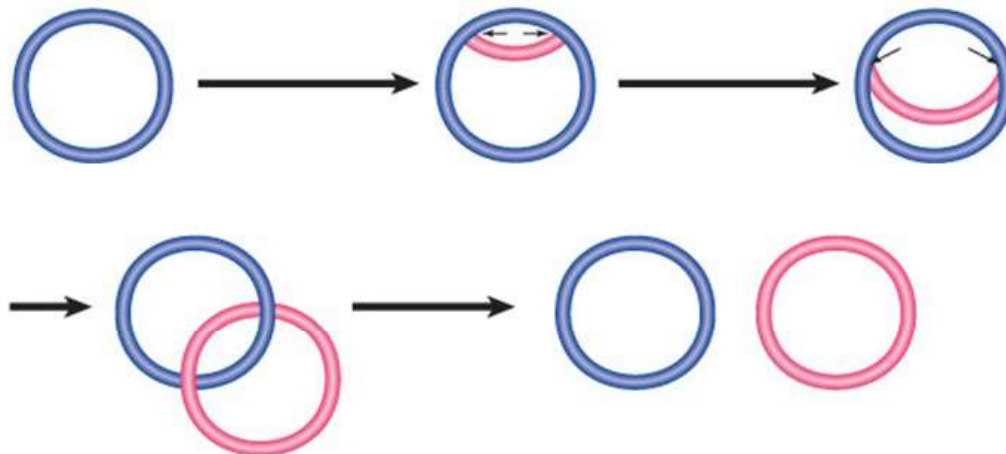
Explain why this is necessary?

Answer:

Linear chromosome-ends look like DNA double strand breaks and may therefore be a substrate for the non-homologous end-joining pathway. Such activity may result in formation of circular chromosomes or chromosome fusions (I hope you answered “D” in MCQ 6). To avoid such undesired events, the single stranded 3'-end of the telomeres invades a neighbouring double-stranded section of the telomeres to form a T-loop. In this way, the end is buried and not accessible for DNA double strand break repair.

Problem B2. (6 %). Replication

Uffe: The figure shows replication of the circular bacterial chromosome.



Explain what enzyme activity is required to separate the two circles and

Answer:

You will need a topoisomerase type II activity, which is able to make a DNA double-strand break on one circle and allow the intact DNA of the other circle to pass through the break. Topoisomerase II is covalently linked to

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describe at the molecular level how the enzyme is able to decatenate the circles.	both end of the break in this process and it is therefore able to seal the break again to complete the process. In this way, the two circles are decatenated.
Consider a mutant strain, which is defective in the activity you proposed in task A. Explain the consequence of this defect in the cell division cycle.	Answer: If the two circular chromosomes remain catenated during cell division, it will cause a chromosome segregation issue. One scenario could be that one cell will receive both circles and one none. Alternatively, one circle will break due to mechanical stress during the attempt to segregate. Elimination of this function is therefore detrimental to cell growth.

Problem B3. (4 %). Recombination

CRISPR technology has revolutionized strain engineering. The key feature of the technology is that the Cas9 DNA endonuclease can be easily programmed to specifically cut virtually any sequence in a genome. Many researchers who work with CRISPR technology in fungi prefer to work with strains where the non-homologous end-joining DNA repair pathway has been eliminated. In the example to the right, <u>Y</u>our <u>F</u>avorite <u>G</u>ene, <i>YFG</i>, is deleted in a CRISPR experiment using a linear DNA fragment containing the selectable marker <i>URA3</i> flanked by homologous sequences up- and down-stream of <i>YFG</i> as repair template delivered by transformation. Selection for the marker gene can be done on solid medium containing uracil as the original strain is missing this gene.	
Explain which repair mechanism in the cell is used to repair the DNA double strand break induced by Cas9 in the example above.	Answer: The cell can use homologous recombination to repair the Cas9 induced break in <i>YFG</i> using the linear fragment as repair template exploiting that it contains a selectable marker flanked by regions homologous to <i>YFG</i>. The resulting crossovers incorporate the marker at the expense of <i>YFG</i>.

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Considering that random DNA double strand breaks are often, and unavoidable, formed as the result of cosmic irradiation and the presence of reactive oxidative molecules formed by the regular metabolism of the cell; explain why it is an advantage to eliminate the non-homologous end-joining pathway if the goal is to specifically delete a gene in a CRISPR experiment.	Answer: In the presence of the non-homologous end-joining pathway, the linear DNA repair fragment may be inserted into the random non-Cas9 induced breaks. If they do, these events will result in transformants that can grow on the selective medium, but where <i>YFG</i> is not deleted. Hence, undesired background will be formed.
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Problem B4. (4 %). The central dogma

Consider a eukaryotic mRNA encoding a protein of 333 amino acids. The length of the mature mRNA was found to be 1350 bases. An analysis of the DNA sequence suggested the presence of two introns with a length of 500 and 200 bases, respectively. Assume that the size of the polyA tail is 250 bases and the polyadenylation site is 50 bases after the stop codon.	
How many exons does the gene encompass?	3
Calculate the approximate length of the leader	Answer: 50 bases
Calculate the approximate length of the pre-mRNA	Answer: 1800 bases
Which experimental method could have been used to measure the length of the mature mRNA (name only)	Answer: Northern blot

Problem B5. (4 %). Transcription

In a study of the constitutive, prokaryotic promoter, three different mutants with reduced transcription levels were isolated. The relative mRNA and the DNA sequence in the three mutants and wildtype were determined. In the figure, the sequences were aligned, and the first bases in the sequence were arbitrarily set to 1. The identified mutations are shown by underscoring. The relative mRNA levels in the different strains are shown at the right.

	1	10	20	30	40	50	Rel. mRNA
WT	CCGTATGCTTTGACAGCTTCACTGATGCCTGGTATAATCTGGACTAGGATCTCC						100
M1	CCGTATGCT <u>CC</u> GACAGCTTCACTGATGCCTGGTATAATCTGGACTAGGATCTCC						1
M2	CCGTATGCTTTGACAGCTTCACTGATGCCTGG <u>CG</u> TAAATCTGGACTAGGATCTCC						1
M3	CCGTATGCTTTGACAGCTTCACTGATGCCTGGTATAATCTGGACTGGGATCTCC						50

Propose a method that could have been used to determine the mRNA levels (Name only)	Answer: RT-PCR or construction of a transcriptional fusion to a reporter gene or RNA seq
Write the first five bases in the wildtype mRNA initiated at the promoter	5'-AGGAU

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The low expression observed in Mutant M1 and M2 is believed to be caused by effects on two different steps in the initiation process. Propose which.	M1: The mutation is in the -35 – hence RNA polymerase binding is most likely affected. M2: The mutation is in the -10 – hence open complex formation (isomerization) is most likely reduced
How can the reduced expression level in M3 be explained?	It is the starting nucleotide in RNA synthesis that is changed from A to G. In most promoters A is the preferred start nucleotide. The G is less efficient – hence a lower expression level.

Problem B6. (4 %). Translation

Describe briefly the role of the three listed protein factors in eukaryotic translation initiation	
PABP	The PolyA-Binding Protein PABP binds to the polyA tail and is required for the formation of the Cap-binding complex. By interaction with eIF4G
eIF4E	eIF4E is part of the Cap binding complex and is responsible for specific binding to the 5'-Cap. After formation
eIF3	Prevent the large and small ribosomal subunits to interact

Problem B7. (4 %). Regulation

Explain the role of a boundary element (insulator) in eukaryotic gene regulation and chromatin remodeling.
Answer: Boundary elements prevent the spread of chromatin modifications or structural changes from one genomic region to another. They are indispensable to keep the heterochromatin/Euchromatic structure since histone remodelling-complexes are not DNA sequence-dependent and need the boundary elements to stop modifying the histones.

Problem B8. (4 %). Virus

What is the difference between the multiplicity of infection (MOI) and the burst size of a certain bacteriophage?	Answer: The multiplicity of infection (MOI) refers to the number of viral particles produced per host cell, while the burst size represents the number of virions released per infected cell.
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How is the multiplicity of infection (MOI) and the burst size of a bacteriophage calculated?	The MOI is determined by dividing the number of viral particles by the number of host cells. The burst size is calculated through a one-step growth experiment. After infecting a bacterial culture, samples are collected at different time intervals, and the plaque-forming units (PFUs) are measured. The burst size is calculated by dividing the maximum number of PFUs attained in equilibrium by the PFUs obtained during the latency period.
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Part C. The (36 %). The Histidine Biosynthesis Operon (His Operon) in *Escherichia coli*

The Histidine Biosynthesis Operon (*his* operon) in *Escherichia coli* controls the production of the amino acid histidine. This operon consists of a set of genes involved in histidine biosynthesis from precursor molecules. The regulation of this operon is mainly based on the availability of histidine within the cell.

The structure of the operator is as shown in the genetic map (fig 1).

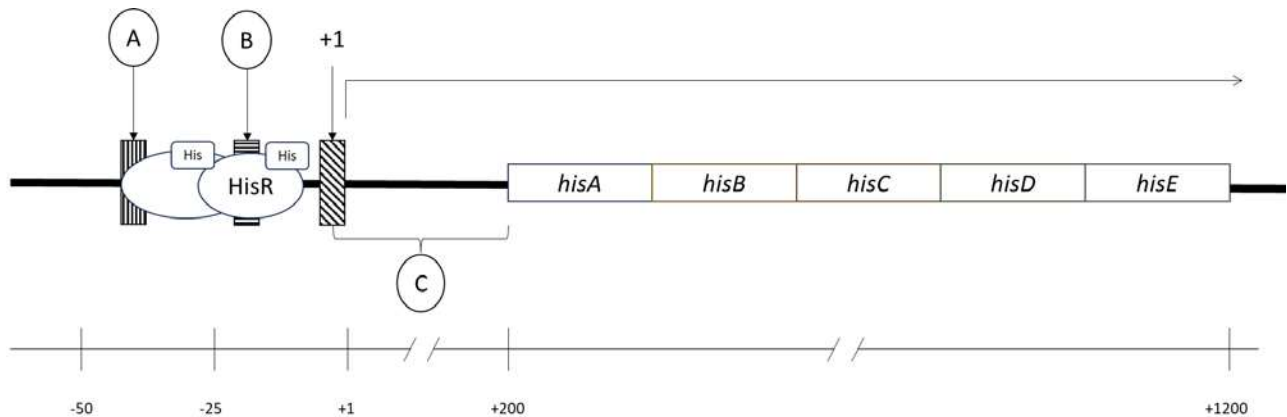


Figure 1. The genetic map of the *his* operon. The upper arrow symbolizes the RNA produced from the operon, while the lower line serves as the reference ruler, designating +1 as the initiation point of transcription. The genes *hisA*, *hisB*, *hisC*, *hisD*, and *hisE* encode enzymes—HisA, HisB, HisC, HisD, and HisE—critical for the synthesis of histidine. Additionally, HisR, a protein encoded by the gene *hisR* situated elsewhere in the bacterial genome, demonstrates an affinity for binding to the promoter of the His Operon. HisR exhibits binding capabilities to the amino acid Histidine and engages with the DNA in a dimeric structure. A, B, and C are addressed in the questions below.

Question 1 – (2%). Regarding its function, how would you classify the genes *hisA*, *hisB*, *hisC*, *hisD*, *hisE*, and *hisR*.

Answer:

hisA, *hisB*, *hisC*, *hisD*, *hisE*, encodes for enzymes and therefore are **structural genes** while *hisR* is a **regulatory gene**.

Question 2 – (4%). Why is an operon structure possible in bacteria and not eukaryotes?

Answer:

Translation initiation in bacteria is controlled by the Shine-Dalgarno (SD) sequence. In eukaryotes it is the 5' cap. In bacteria each open reading frame has its own SD sequence.

Question 3 – (4%). What specific DNA sequence characteristic do you anticipate the dimer of HisR to bind to, considering it forms a dimer to bind to the DNA?

Answer:

The sequence is likely to be palindromic.

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Question 3 - (2%) In Figure 1, name the elements A, B, and C.

Answer:

A: -35

B: -10

C: leader sequence

A research team investigated the regulation of the *his* operon by conducting experiments under varying conditions of histidine availability. They studied both wild-type bacteria and a set of mutants in the presence and absence of histidine to understand how the operon's regulation functions. To facilitate the readout, the genes *hisA*, *hisB*, *hisC*, *hisD*, and *hisE*, have been substituted by a *lacZ* gene encoding the enzyme β -galactosidase whose enzymatic activity can be easily measured in the laboratory.

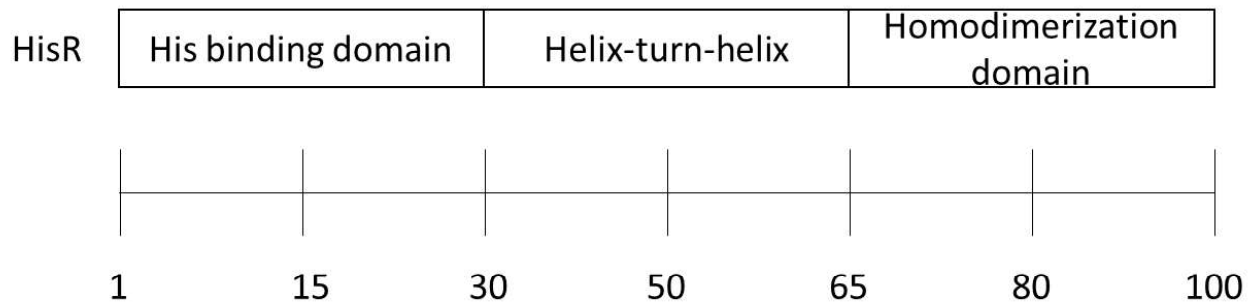


Figure 2. Domain structure of the protein HisR encoded by the gene *hisR*. The numbered scale indicates the number of the amino acids starting at the N-terminal end.

Strain	Mutation	Relative β -galactosidase activity (%)	
		-Histidine	+Histidine
1	None (wild-type)	100	10
2	$\Delta hisR$	100	100
3	$\Delta hisR$ + plasmid- <i>hisR</i>	100	10
4	HisR- Δ 30-65	100	100
5	HisR- Δ 1-30	100	100
6	HisR-L80S	100	50

Table 1. Mutations of HisR. Activity of the β -galactosidase in the presence and the absence of Histidine. **Strain 1** is the wild-type strain, **Strain 2** has a complete deletion of the *hisR* gene, **Strain 3** has a complete deletion of *hisR*, and has been transformed with a plasmid encoding the HisR wild-type, **Strain 4** lacks residues 30 to 65 of HisR, **Strain 5** lacks the first 30 amino acids of HisR, **Strain 6** has a single point mutation in the *hisR* gene in which the Leucine (L) in position 80 has been substituted by a Serine (S).

Question 4 - (4%). What kind of regulatory element (cis/trans) is encoded by *hisR*? Briefly explain your response.

Answer:

A trans-regulatory element. When *hisR* is deleted from the host it can be supplemented in a plasmid, which means it is a diffusible element independent from the regulated genes.

Question 5 – (4%). Examining the β -galactosidase activities found for Strain 1 and Strain 2 from Table 1, how does HisR regulate the expression of the His operon in response to histidine availability?

Answer:

In the wild-type strain, the HisR gene is intact. In the absence of His, the expression is high while in the presence of His the expression is low.

When HisR gene is deleted (strain 2) the expression is constitutively high, which means that HisR is a repressor of the His operon.

In the presence of Histidine, HisR binds Histidine and undergoes a conformational change that increases the affinity for the operator site, which overlaps with the promoter, avoiding the binding of the RNA polymerase and therefore inhibiting the expression of the operon.

Question 6 – (6%). Explain the result obtained in Strains 4, 5, and 6 considering the HisR domains shown in figure 2 affected by the mutations.

Answer:

Strain 4:

lacks the DNA-binding domain (helix-turn-helix) and therefore in the presence or absence of His, it cannot bind to DNA, leading to a high signal regardless of the presence of the amino acid.

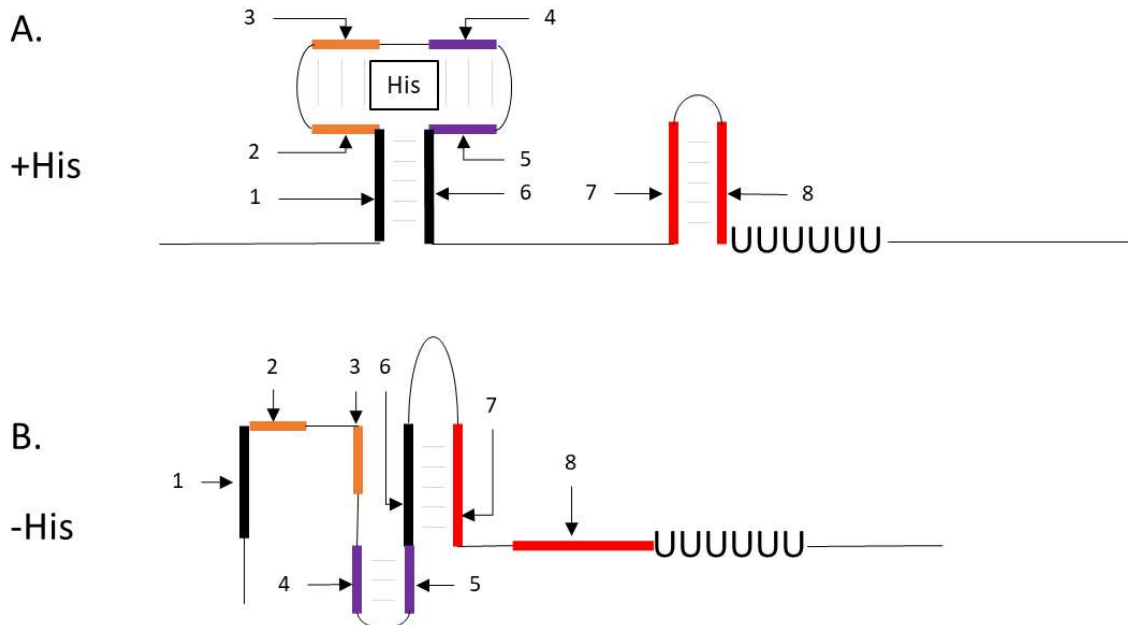
Strain 5:

lacks the Histidine binding domain, preventing interaction with the inducer. Consequently, HisR remains unbound to DNA, enabling operon transcription, leading to a high signal in both Histidine presence and absence.

Strain 6:

retains the ability to bind Histidine and DNA. However, a mutation affects the homodimerization domain. In the presence of Histidine, HisR can bind to DNA but inadequately, failing to effectively prevent operon transcription despite some interaction.

The *his* operon has an extra layer of regulation. By adopting mutually exclusive structures in the mRNA leader sequence in the presence or absence of Histidine as depicted in Figure 3, the transcription of the *his* genes is affected



Course 27026 – Molecular biology – E2023

Figure 3. Mutually exclusive structures adopted by the mRNA leader sequence. This figure illustrates the alternative structures adopted by the mRNA leader sequence in two conditions: (A) in the presence of histidine (His) and (B) in the absence of histidine. Each fragment within the schematics is denoted by an identical number (1-8) in both representations (A and B). The grey parallel lines indicate the formation of hydrogen bonds between corresponding parallel fragments in the structures.

Question 7 - (2%). What would you call the regulatory mechanism illustrated in figure 3 for the His operon?

Answer:

It is a riboswitch/Attenuator

Question 8 - (4%). Which is the common name of the structures formed in Figure 3A and 3B, respectively, and what are the consequences for the expression of the His operon

Answer:

Figure 3A: Terminator. The RNA polymerase detached from the mRNA reducing the transcription of the His operon.

Figure 3B: Antiterminator. The RNA polymerase continues the transcription of the complete mRNA.

To better understand the mechanism of the regulation at the leader sequence level, the research team has produced a series of mutations fused to the β -galactosidase reporter gene in the presence and the absence of His.

Question 9 – (4%). Predict the activity of the reporter gene in the following table.

Answer:

Strain	Leader sequence	Relative β -galactosidase activity (%)	
		-Histidine	+Histidine
1	1-8 (wild-type)	100	10
2	$\Delta 8$	100	100
3	$\Delta 6$	10	10
4	$\Delta 1-6$	10	10
5	$\Delta 7-8$	100	100

Note 1: all the elements of the promoter are wild type, and the modifications affect only the leader sequence.

Note 2: ΔX means that all the fragments are included except for the number X.

Note 3: $\Delta X-Y$ means that all the fragments are included with the exception of the range between X and Y included.

**MC Exam 27026 E19****Side 1**

MC part 1 of 5

☒ Vis rigtige svar
☐ Skjul rigtige svar**Spørgsmål 1**

Fill in the right words in this sentence: In _____ blot, RNA is determined, in _____ blot, DNA is determined, whereas _____ blotting is used to quantify proteins.

- ☒ Northern, Southern, Western.
- ☐ Northern, Western, Southern.
- ☐ Southern, Northern, Western.
- ☐ Western, Northern, Southern.
- ☐ Western, Southern, Northern.

Spørgsmål 2

Which of the following methods **cannot** be used to introduce a DNA molecule into a cell?

- ☒ Centrifugation.
- ☐ Liposome fusion with target cell.
- ☐ Microinjection into the nucleus.
- ☐ Transformation after chemical treatment.
- ☐ "Gene gun" ie nanospheres coated with DNA.

Spørgsmål 3

A newly replicated eukaryotic chromosome contains numerous small patches of RNA. This is because _____?

- ☒ RNA is used a primer for synthesis.
- ☐ NTPs are often accidentally used for synthesis instead of dNTPs.
- ☐ mRNA is often incorporated into DNA during replication.
- ☐ tRNA is often incorporated into DNA during replication.
- ☐ rRNA is often incorporated into DNA during replication.

Spørgsmål 4

The role of a DNA helicase during replication is to _____?

- ☒ separate DNA strands of a chromosome prior to replication.
- ☐ maintain the DNA helix as a duplex after replication.
- ☐ neutralize DNA supercoils during replication.
- ☐ synthesize short helical stretches of RNA prior to replication.
- ☐ establish the covalent bonds between the Okazaki fragments.

Side 2

MC part 2 of 5

Spørgsmål 5

Pyrimidine dimers are formed as the result of UV irradiation. These lesions are in eukaryotes and in the dark, these lesions are typically removed by ____?

- ☒ nucleotide excision repair system.
- ☐ mismatch repair system.
- ☐ base excision repair system.
- ☐ non-homologous end-joining pathway.
- ☐ trans lesion synthesis.

Spørgsmål 6

In *E. coli*, hemimethylated DNA is used to increase the fidelity of the mismatch repair system as it allows the enzymes to ____?

- ☒ distinguish newly synthesized DNA strand from the old template strand.
- ☐ cleave the methylated DNA strand more easily.
- ☐ detect a mispaired basepair.
- ☐ replicate the damaged strand after the lesion has been.
- ☐ avoid mismatch formation in the first place.

Spørgsmål 7

Rolling circle replication is used ____?

- ☒ by some circular DNA parasites to make many copies after replication initiation.
- ☐ to amplify the number of rDNA repeats.
- ☐ by some prokaryotes to avoid use of hemimethylated DNA.
- ☐ by some transposons as part of integrating into another DNA strand.
- ☐ to amplify the number of telomeres.

Spørgsmål 8

In front of the progressing replication fork, DNA polymerases always generate ____?

- ☒ positively supercoiled DNA.
- ☐ negatively supercoiled DNA.
- ☐ relaxed DNA.
- ☐ broken DNA.
- ☐ formation of telomeres.

Side 3

MC part 3 of 5

Spørgsmål 9

DNA polymerase interacts with the clamp protein. This enhances _____?

- ☒ processivity of DNA replication.
- ☐ DNA repair fidelity.
- ☐ DNA replication fidelity.
- ☐ removal of positive supercoils.
- ☐ removal of negative supercoils.

Spørgsmål 10

Fill in the right words in the sentence: The prokaryotic RNA polymerase binds initially to the _____ by a _____ in the _____. Then the DNA _____ in the _____, and the _____ is stabilized by interaction between _____ and the _____.

- ☒ - 35 sequence, Helix-Turn-Helix motif, sigma factor, denatures, - 10 region, open complex, single stranded DNA, sigma factor.
- ☐ - 10 sequence, Ssb motif, sigma factor, denatures, - 35 region, open complex, the -35 sequence, sigma factor.
- ☐ - 35 sequence, Helix-Turn-Helix motif, ribosome, denatures, - 10 region, open complex, single stranded DNA, ribosome.
- ☐ - 10 sequence, Ssb motif, sigma factor, denatures, - 35 region, open complex, the -35 sequence, sigma factor.
- ☐ - 35 sequence, DNA binding motif, ribosome, denatures, - 10 region, open complex, single stranded DNA, ribosome.

Spørgsmål 11

Fill in the right words in the sentence: The transcription initiation of eukaryotic RNA polymerase II promoters is initiated by binding of _____ to the promoter region. Subsequently, the _____ and additional _____ are recruited. The _____ of RNA polymerase II is _____ facilitating the entry into elongation mode.

- ☒ TF_{II}D, RNA polymerase II, Transcription Factors, CTD, phosphorylated
- ☐ RNA polymerase II, TF_{II}D, Transcription Factors, CTD, phosphorylated
- ☐ RNA polymerase II, CTD, Transcription Factors, TF_{II}D, phosphorylated
- ☐ TF_{II}D, Ribosome, Transcription Factors, CTD, acetylated
- ☐ RNA polymerase II, TF_{II}D, Transcription Factors, CTD, acetylated

Spørgsmål 12

What is the most abundant initial step in eukaryotic mRNA degradation?

- ☒ Deadenylation.
- ☐ Decapping.
- ☐ RNase E cleavage.
- ☐ 3'-5'-exonuclease degradation.
- ☐ 5'-3'-exonuclease degradation.

Side 4

MC part 4 of 5

Spørgsmål 13

How is the accuracy of translation improved in tRNA charging?

- ☒ By excluding smaller and larger amino acids from the active site of the synthetases.
- ☐ By ATP hydrolysis.
- ☐ By specific proof-reading enzymes.
- ☐ The ribosome has a proofreading site.
- ☐ Wrong Aminoacyl-tRNAs does not fit in the A site of the ribosome.

Spørgsmål 14

The ribosomal E site ____?

- ☒ transiently binds deacylated tRNA.
- ☐ binds aminoacyl-tRNA.
- ☐ binds the peptidyl-tRNA.
- ☐ binds the Shine–Dalgarno sequence.
- ☐ binds the mRNA start codon.

Spørgsmål 15The flux through a metabolic pathway is subject to control. Which of the following control mechanism is **not** valid?

- ☒ Transposition.
- ☐ Allosteric control of the enzymes in the pathway.
- ☐ Regulation of transcription initiation.
- ☐ Covalent modification of histones.
- ☐ Regulation of translation initiation.

Spørgsmål 16

Which of the following statements about CRP is false?

- ☒ It binds to the *lacO1* operator.
- ☐ It is a dimer.
- ☐ It is activated by a single molecule of cAMP.
- ☐ It is activated when cellular glucose levels are low.
- ☐ It facilitates RNA polymerase binding to the promoter.

Side 5

MC part 5 of 5

Spørgsmål 17

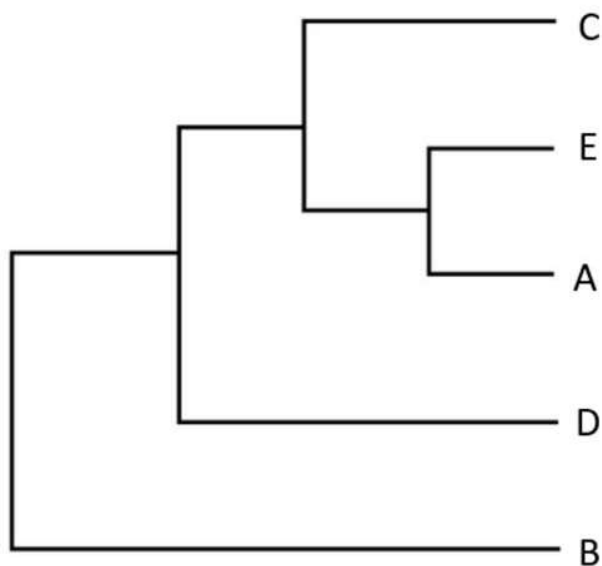
In the context of phage infection, “Induction” means ____?

- ☒ the transition from a lysogenic to a lytic state.
- ☐ promotion of the expression of phage genes.
- ☐ host cell lysis.
- ☐ integration of phage DNA into the bacterial chromosome.
- ☐ packaging of phage DNA into protein coats.

Spørgsmål 18

sRNA generally functions as a regulator by ____?

- ☒ base-pairing with a single-stranded region.
- ☐ influencing ribosomal activity.
- ☐ recruiting proteins to gene promoters.
- ☐ catalyzing biochemical reactions.
- ☐ participating in chromatin remodeling.

Spørgsmål 19

Based on the phylogenetic tree below, which statement is true?

- ☒ Species C is more closely related to species D than to species B.
- ☐ Species C is more closely related to species B than to species A.
- ☐ Species A is more closely related to species C than to species E.
- ☐ Species A is more closely related to species B than to species D.
- ☐ Species B is more closely related to species A than to species E.

Spørgsmål 20

Genetic drift is ____?

- ☒ independent on selection.
- ☐ the principle that lead to codon wobble.
- ☐ independent on spontaneous mutation.
- ☐ only observed during “Adaptive Laboratory Evolution” experiments.
- ☐ independent on selection.



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Multiple Choice exam 27026 10-12-2020

Side 1

Group 1

☒ Vis rigtige svar
☐ Skjul rigtige svar

Spørgsmål 1

What is measured in a 1) Western blot, 2) Southern blot, and 3) Northern blot?

- ☐ 1) Protein, 2) RNA, 3) DNA
- ☒ 1) Protein, 2) DNA, 3) RNA
- ☐ 1) RNA, 2) Protein, 3) DNA
- ☐ 1) DNA, 2) Protein, 3) RNA
- ☐ 1) DNA, 2) RNA, 3) Protein

Spørgsmål 2

The centromere is used to _____?

- ☐ position the nucleus in the center of a eukaryotic cell
- ☒ segregate sister chromatids during cell division
- ☐ organize the assembly of a replication fork
- ☐ position RNA polymerase at the center of a TATA box
- ☐ position the spliceosome during intron splicing

Spørgsmål 3

Histone Acetylation is used to _____?

- ☒ regulate gene transcription in eukaryotes
- ☐ regulate gene transcription in prokaryotes
- ☐ regulate RNA splicing
- ☐ regulate translation of RNA
- ☐ regulate reverse transcriptase activity in the life cycle of a retrovirus

Spørgsmål 4

Replication of eukaryotic chromosomes depend on RNA because ____?

- ☐ active gene transcription is directly required for replication initiation
- ☐ replication initiation takes place at promoters
- ☐ replication initiation takes place at the non-transcribed strand during transcription
- ☐ mRNA is used as primers for DNA replication
- ☒ DNA polymerase cannot initiate DNA synthesis without a primer

Spørgsmål 5

Okazaki fragments occur during _____?

- ☐ degradation of viral DNA
- ☐ degradation of mRNA
- ☐ telomerase activity
- ☐ during leading strand DNA synthesis
- ☒ during DNA lagging strand synthesis

Side 2

Group 2

Spørgsmål 6

Non-homologous end-joining is used to _____?

- ☐ attach telomeres to chromosome ends
- ☒ repair a DNA double strand break
- ☐ splice RNA molecules during removal of introns
- ☐ seal nicks between Okazaki fragments
- ☐ insert DNA into a specific gene by homologous recombination

Spørgsmål 7Correction of replication errors by the *E. coli* mis-match repair system is facilitated by _____?

- ☐ the Okazaki fragments
- ☐ the presence of introns
- ☐ the superhelicity of DNA
- ☐ trial and error
- ☒ the presence of hemi-methylated DNA

Spørgsmål 8

The F plasmid is able to transfer between _____?

- ☐ a phage particle and an *E. coli* cell
- ☐ a viral particle and a human cell
- ☒ two *E. coli* cells
- ☐ an *E. coli* cell and a human intestine cell
- ☐ two human cells

Spørgsmål 9

In the retroviral life-cycle, tRNA is used to make retroviral DNA from RNA by _____?

- ☒ acting as primer
- ☐ inhibiting integrase
- ☐ inhibiting RNases
- ☐ inhibiting DNases
- ☐ inhibiting reverse transcriptase

Spørgsmål 10

Transcription initiation in prokaryotes require _____?

- ☐ a primer
- ☐ interaction between the -35 sequence in the promoter DNA and the core RNA polymerase
- ☒ formation of the open complex
- ☐ transcription factor TF_{IID}
- ☐ an upstream activating sequence (UAS)

Side 3

Group 3

Spørgsmål 11

Class U2 dependent introns ____.

- ☒ have the consensus sequence GU at the 5' end of the intron
- ☐ are found abundantly in mitochondrial encoded genes
- ☐ are self-splicing introns
- ☐ are part of the codon sequence
- ☐ have their recognition sequence inside the exons

Spørgsmål 12

Which enzymes are NOT involved in mRNA degradation?

- ☒ Restriction endonucleases
- ☐ 5'-3'- exonucleases
- ☐ 3'-5'- exonucleases
- ☐ Phosphatases
- ☐ Deadenylases

Spørgsmål 13

Which factor is required for loading of the aminoacyl-tRNA into the eukaryotic ribosome?

- ☐ EF-TU
- ☒ eEF1a
- ☐ No additional factor is needed
- ☐ GTP
- ☐ ATP

Spørgsmål 14Consider the following Anticodon: 3'-GCI-5'. Which amino acid will be inserted during translation?

- ☐ Valine
- ☐ Alanine
- ☐ Proline
- ☐ Isoleucine
- ☒ Arginine

Spørgsmål 15

Which of the following molecules in connection with prokaryotic regulatory systems is a metabolite?

- ☐ A repressor
- ☐ An activator
- ☐ An operator
- ☐ An operon
- ☒ A co-activator

Side 4

Group 4

Spørgsmål 16

Which statement addressing eukaryotic gene regulation is wrong?

- ☐ A true activator makes direct contact with the basal apparatus
- ☐ An activator may work by preventing a repressor to bind to its target
- ☐ Both activators and repressors may be covalently modified
- ☒ A repressor prevents transcription by degradation of TF_{IID}
- ☐ Both activators and repressors are gene encoded

Spørgsmål 17

Which statement addressing regulation by small RNAs is wrong?

- ☐ cis encoded antisense sRNA always have a perfect match with the target
- ☐ Trans-encoded antisense sRNA often require the Hfq chaperone for activity
- ☒ sRNA facilitates binding of an attenuator
- ☐ sRNAs can functions as regulators by base pairing to specific target RNAs
- ☐ The kissing complex is the initial contact between an antisense RNA and its target

Spørgsmål 18

On which types of viruses do antiviral drugs act that inhibit the viral DNA polymerase?

- ☐ DNA and RNA viruses
- ☐ RNA viruses
- ☐ RNA and Retroviruses
- ☐ DNA, RNA and retroviruses
- ☒ DNA and retroviruses

Spørgsmål 19

A suspension of virulent phages is added to a bacterial culture. What will happen to the bacteria?

- ☒ All cells will lyse
- ☐ Some cells will lyse, some cells will continue to grow
- ☐ All bacterial cells will continue to grow
- ☐ Only cells where the phage integrate into the chromosome will survive
- ☐ Any of the suggested outcomes could happen

Spørgsmål 20

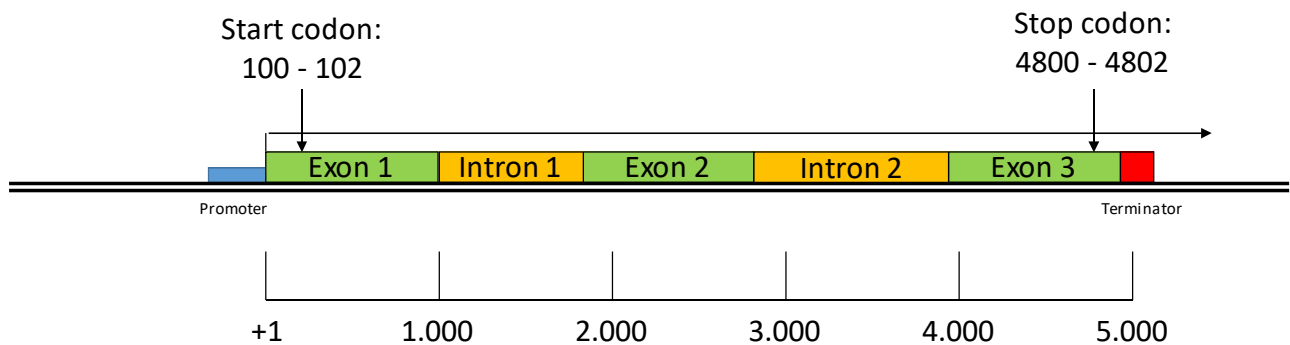
“When several nucleotide triplets encode the same amino acid, one tends to be used in genes more frequently”, what is this observation called

- ☐ mutational bias
- ☐ genetic drift
- ☒ codon bias
- ☐ synonymous substitution
- ☐ gene conversion bias

Example Exam 27026 B part.

B.1.

Below is a schematic of gene Y, which encodes protein Y. The promoter region is indicated by the blue box. Transcription begins at +1, and the transcribed RNA is shown.



How long is the pre-mRNA? Two different transcripts are produced from this gene, one is approximately 2000 nucleotides long, the other is approximately 3000 nucleotides long. Explain how two different transcripts can be produced from this gene. Assume each transcript produces a protein. What is the approximate size of the two proteins produced from this gene? Briefly describe how the transcription may initiated. Include the different enzymes and proteins involved in the initiation process.

The pre-mRNA is approximately 5000 nucleotides. The two different mature mRNA's are obtained by alternative splicing. In the 3000 bases transcript all three exons are present, whereas the 2000 b transcript misses the middle exon, as the start and stop codon is found in exon 1 and 3, respectively. The amino acids made from the two mRNA are $900/3 + 1000/3 + 800/3 = 900$ and $900/3 + 800/3 = 566$ amino acids – approximately. Being a “normal” gene, the transcription is driven by RNA pol II. Here the transcription initiation process is started by assembly of the $TF_{II}D$ at the promoter. The RNA pol II is recruited together with $TF_{II}B$ and $TF_{II}F$. Subsequently $TF_{II}E$ and $TF_{II}H$ become associated with the other elements in the Basal apparatus at the promoter, and transcription starts by melting of the DNA. To ensure promoter clearance the C-terminal domain of the RNA pol II is phosphorylated, the different transcription factors dissociates, at transcription is initiated.

B.2.

You have isolated a new virus, and the virions was shown to contains RNA. To find out how the virus works, you infect host cells with this virus and discover that upon infection, the host cell makes lots of viral RNA molecules and various viral proteins. What type of virus have you discovered, and describe how new virus particles are formed after infection. Include both how the capsid and genome are produced.

It is most likely a Retro virus. After infection, the RNA genome is converted to DNA by the enzyme Reverse Transcriptase, and double stranded DNA is made by the aid of the host DNA polymerase. The DNA may be integrated in the host chromosome, and start making mRNA for proteins – both structural and regulatory. Also genomic RNA is made, and packed into virus particles together with the Reverse Transcriptase enzyme.

B.3.

Describe how Southern and Northern Blots are performed, and what kind of molecules that are determined in the two experiments.

Southern blot is used to identify specific DNA: Purified DNA is digested with a restriction enzyme, and separated based on size by gel electrophoresis. The DNA is transferred to a membrane by blotting, and a labelled probe is used to identify specific DNA on the gel by base pairing. Excess probe is removed by washing, and the bands that binds the probe is identified by the property of the label.

Northern blot is used to identify specific RNA: Purified RNA is separated based on size by gel electrophoresis. The RNA is transferred to a membrane by Blotting, and a labelled probe is used to identify specific RNAs on the gel by base pairing. Excess probe is removed by washing, and the bands that binds the probe is identified by the property of the label.

B.4. Describe in broad terms how a riboswitch can control gene expression at the level of translation in a prokaryote.

A riboswitch binds a metabolite and changes the secondary structure. In one of the structures, the Shine-Delgarno sequence is sequestered in a double stranded structure denying the ribosome access; hence, translation is prevented. In the alternative conformation, the Shine-Delgarno sequence is exposed as a single stranded sequence to the Ribosome, and thus being translated.

Regulation of the galactose degradation genes in *Aliivibrio fischeri*.

Beta-D-galactose is converted to alpha-D-glucose-1-phosphate in two enzymatic steps catalyzed by two enzymes, galactose mutase (GalM) and galactokinase (GalK). The enzymes are encoded by the genes *galM* and *galK*. The two genes are organized in an operon (Figure 2.1).



Figure 2.1. Genetic organization of the *galMK* operon in *A. fischeri*. The arrow indicates the extent of the transcribed mRNA.

Following transposon mutagenesis, two mutants (UT11 and UT12) were isolated. The mutation in UT12 was found to be close to the promoter of the *galMK* operon with genetic mapping, while the mutation in UT11 was identified to be in a regulatory gene (*galR*) with a great distance to the *galMK* operon.

In a study of the mutants, the content of *galMK* mRNA was quantified in the wild type strain (wt) and in each of the two mutants. RNA was extracted from each of the strains after growth in medium with either glucose, galactose, or with a mixture of glucose and galactose as the sole carbon source.

It was already known that the *galMK* operon is expressed from a cAMP/CRP activated promoter that is induced 100 times when no glucose is present. Table 2.1 shows the relative mRNA amounts in the four strains.

Strain	wt			UT11			UT12		
Addition	glucose	galactose	glu + gal	glucose	galactose	glu + gal	Glucose	galactose	glu + gal
Relative amount of mRNA	1	500	5	5	500	5	1	5	5

Table 2.1. *galMK* mRNA levels in *A. fischeri* wild type and mutants. The cells were grown in medium with either galactose, glucose or both glucose and galactose (glu + gal). The cells were harvested and the *galMK* mRNA levels determined.

Question 2.1 (3 points). What effect does the addition of galactose have on the expression of the *galMK* operon in the wild type when there is also glucose present?

It induces the expression of *galMK* 5 times

Question 2.2 (3 points). Why is the induction 500 times when the cells grow on galactose when compared to glucose?

The 500 times induction on galactose is due to the fact that there is no catabolite repression (100 times), but also galactose induction, which induces 5 times. That is $100 * 5 = 500$

Question 2.3 (5 points). Based on information on the location of the mutation in UT11 and the expression levels of the *galMK* genes in the table, provide a suggestion on how *galMK* can be regulated and what role galactose plays.

In the presence of glucose there is catabolic repression - that is, low expression with glucose. The operon is also regulated by a repressor where galactose is an inducer. In UT11, the repressor is inactive. Therefore, the level is higher with glucose relative to the wild type. There is 20 times of repression when the repressor is present - but only in the absence of galactose.

Question 2.4 (3 points). How can the phenotype of UT12 be explained?

In the presence of glucose there is catabolite repression. In UT12 there is no effect of glucose. Levels are dependent on galactose. One can imagine that the cAMP-CRP activator binding site is destroyed, therefore there is no activation of the expression.

Question 2.5 (3 points). If a limited amount of glucose was added in a mixture with galactose, one would observe that the expression in the wild type reached 200 at the end of the experiment. Explain why.

In the presence of glucose there is catabolic repression. This means that glucose is consumed first. Once the cell has metabolized all the glucose, catabolite repression is relieved, and *galMK* will be expressed.

Regulation of tetracycline resistense

Genes coding for resistance to the antibiotic tetracycline are very useful as selection markers in transformations and are therefore very important in biotechnology research. A very effective gene (*tetA*) was found that gives tetracycline resistance in the bacterium *Salmonella enterica*.

A biotech company would make a cloning vector for *Escherichia coli* with the resistance gene. Although it was easy to transform *E. coli* with the plasmid, it was very difficult to use the tetracycline resistance marker for selection as very few transformants could survive on tetracycline plates.

It was found that if *E. coli* cells were allowed to grow with low non-toxic (sublethal) concentrations of tetracycline (0.1 µg / ml) for one hour, they would subsequently be fully resistant and could grow on plates with tetracycline concentrations in excess of 20 ug/ml. To find the reason for this, the sequence of the cloned DNA fragment was determined. The map is shown in Figure 2.1. During the annotation of the sequence, two genes (*tetA* and *tetR*) were found, which were transcribed in opposite direction. By comparison with known genes, a motif in the *tetR* gene which suggested that the TetR protein could bind a tetracycline molecule.

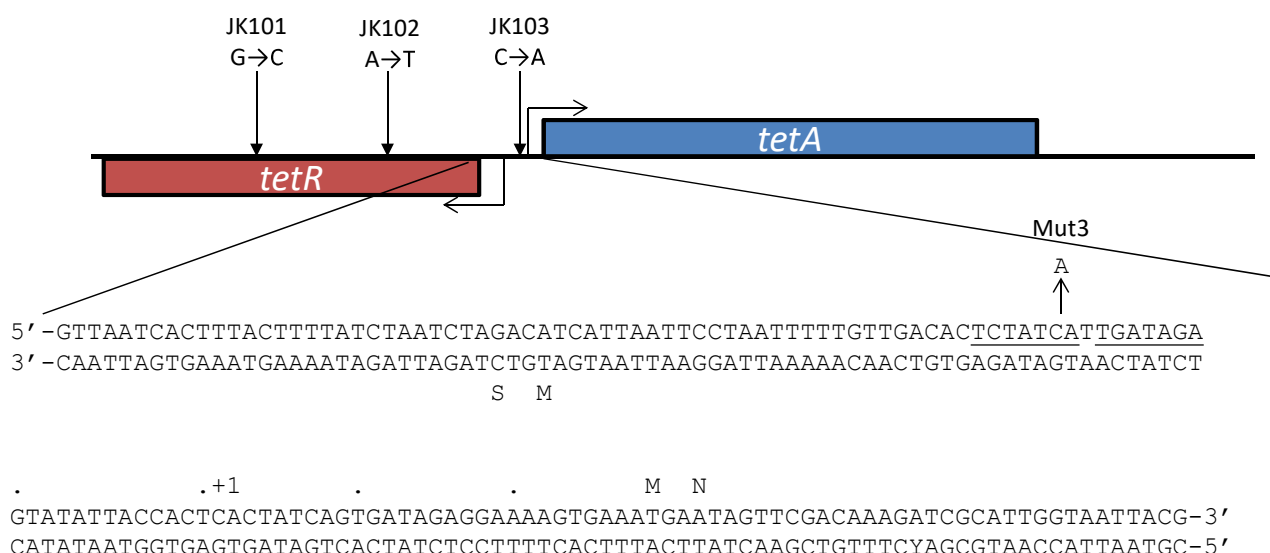


Figure 2.1. Organization of the *tetA* / *tetR* site from *S. enterica*. (Top) The promoter in front of the genes is shown with bent arrows that symbolize the transcription start point (+1). The location of mutations in strains JK101, JK102 and JK103 is indicated and shown with arrows. (Bottom) The sequence of both DNA strands from the area between the two open reading frames. "+1" indicates the first nucleotide in the *tetA* mRNA. The first two amino acids in each of the proteins TetA and TetR are shown. The mutation in the strain JK103 (Mut3) is shown indicating the base change on the top DNA strand.

The Standard Genetic Code

	T			C			A			G		
T	TTT	Phe	F	TCT	Ser	S	TAT	Tyr	Y	TGT	Cys	C
	TTC			TCC			TAC			TGC		
	TTA	Leu	L	TCA			TAA	STOP		TGA	STOP	
	TTG			TCG			TAG			TGG	Trp	W
C	CTT	Leu	L	CCT	Pro	P	CAT	His	H	CGT	Arg	R
	CTC			CCC			CAC			CGC		
	CTA			CCA			CAA	Gln	Q	CGA		
	CTG			CCG			CAG			CGG		
A	ATT	Ile	I	ACT	Thr	T	AAT	Asn	N	AGT	Ser	S
	ATC			ACC			AAC			AGC		
	ATA			ACA			AAA	Lys	K	AGA	Arg	R
	ATG	Met	M	ACG			AAG			AGG		
G	GTT	Val	V	GCT	Ala	A	GAT	Asp	D	GGT	Gly	G
	GTC			GCC			GAC			GGC		
	GTA			GCA			GAA	Glu	E	GGA		
	GTG			GCG			GAG			GGG		

Question 3.1. Write the first five amino acids in TetR and TetA, indicating the N and C terminals.

TetR: N -Met-Ser-Arg-Leu-Asp-C

TetA: N -Met-Asn-Ser-Ser-Thr-C

Question 3.2. Write the sequence of the *tetA* promoter, and underline the most likely -10 and -35 sequences.

TTGACACTCTATCATTGATAGAGTATATTACCACTCACTATCAGT
-35 -10

The inserted DNA fragment in the plasmid was investigated by analyzing mutants obtained by UV mutagenesis. Three different mutants (JK101, JK102 and JK103) were isolated. For the further analysis, a plasmid (pJK999) was constructed which contained only the wild type *tetR* gene. The strain of wild-type *tetA tetR* region was transformed with pJK999. Similarly, pJK999 was introduced into each of the strains JK101, JK102 and JK103. All strains with and without pJK999 were analyzed for growth on high concentration tetracycline plates. In one series of experiments, the bacteria were plated directly onto tetracycline plates. In the second part, they were first incubated for one hour with 0.1 µg/ml sublethal tetracycline concentrations, after which they were plated onto tetracycline plates. The results are shown in Table 1.

Table 1. Growth (+) of strains on agar plates with tetracycline (20 µg/ml)

Strain	Wildtype		JK101		JK102		JK103	
		pJK999 ^{*)}		pJK999 ^{*)}		pJK999 ^{*)}		pJK999 ^{*)}
With pre-incubation ^{**)}	+	+	+	+	-	-	+	+
Without pre-incubation	-	-	+	-	-	-	+	+

^{*)}: pJK999 expresses *tetR* but not *tetA*

^{**)}: Growth with sublethal concentrations of tetracycline (0,1 µg/ml) for one hour

Question 3.3. Describe the effect of the mutation in JK101 and give your suggestion on what type of mutation can explain the observed effect.

It is a loss-of-function mutation that causes the TetR repressor to not repress the expression of *tetA*

Question 3.4. Describe the effect of the mutation in JK102 and give your bid what type of mutation can explain the observed effect.

It is a mutation that causes tetracycline to not bind to TetR, why TetR will repress the expression of *tetA* although induced with tetracycline

Question 3.5. Describe the effect of the mutation in JK103 and give your bid what type of mutation can explain the observed effect.

It is an operator mutation that causes the TetR repressor to not repress the expression *tetA*